

# Weeks 2-10

## Facilitator guides

Preparation, pacing, hints, proofs, source notes, and extra-page solutions.

Week 2 is one shared collection; grade bands below apply to Weeks 3-10.

Give one page at a time. The packet is a menu, not a checklist for one hour.

Each student set includes the same Week 2 library. Print its chosen pages once.

| Week and investigation           | PDF pages |
|----------------------------------|-----------|
| 02 Switches and lamps            | 2-17      |
| 03 Repeating secret machines     | 18-23     |
| 04 Around the ring               | 24-28     |
| 05 Tower cities                  | 29-33     |
| 06 Code-breaking with guarantees | 34-39     |
| 07 Take-away games               | 40-44     |
| 08 Rooks and piles               | 45-50     |
| 09 Bouncing paths                | 51-56     |
| 10 Bridges and delivery routes   | 57-62     |

Print US Letter, single-sided, at 100% / Actual Size.

Page ranges above count this cover as page 1. Bookmarks open each week.

Weekly footers retain their own page numbers and activity IDs for the use log.

Unpiloted. Week 3 updated September 30; other edition IDs remain in footers.

# One collection, many good routes

42 student pages / 50 problems. Prepared September 28, 2026; unpiloted. Choose a few pages. Nobody is expected to work through the book in order.

**Start together on student page 1.** After a child can make and undo a legal move, choose exploration (pp. 2-8), collecting pictures (pp. 9-16), or improving solutions (pp. 17-28). Drawing (pp. 29-34) is always available. Networks (pp. 35-42) are reserve investigations.

## Three stable tables; pages can travel

Ten children: KK1 / 3333 / 445. The non-mathematician parent initially anchors the three youngest children; the other mathematician anchors the four third graders; the organizer anchors the remaining three. This is a staffing arrangement, not a restriction on tasks. Keep seats and adults stable while offering suitable pages across the whole collection. There are no required station rotations.

## Materials and manageable printing

Pencils and erasers, or whiteboard tablets with markers and erasers; ordinary pens work for fresh drawings. Existing two-sided counters are optional. Print ten copies of student p. 1. Suggested reserve stacks: **parent:** three each of pp. 2, 3, 30, 31; **mathematician A:** four each of pp. 9-12 and 15; **mathematician B:** three each of pp. 17-21. These are available choices, not assigned piles. Copy reserve pages before the meeting. Keep one complete master collection to display or sketch from.

## What each adult needs

All three adults: this page and the page finder (guide p. 2). Parent: guide pp. **3-5 and 11-13** give scripts and answers for all exploration/drawing choices. Mathematicians: guide pp. 6-10 and 14-16 supply the remaining keys and proofs. All answer references use the shared student problem numbers.

## A flexible hour

**0-5:** handle materials; draw and erase. **5-10:** demonstrate, then replay one legal move together. **10-30:** selected tasks, solo or shared. **30-35:** stand and stretch. **35-50:** continue, change difficulty, or draw rooms. **50-55:** show one discovery. **55-60:** tidy. End a route after a satisfying attempt; explanation and proof can be a conversation.

## When someone needs more help

Keep the child on a familiar board or drawing task while the adult helps a neighbor. If the parent needs mathematical support, the organizer first asks the other mathematician to cover the two older tables, then visits briefly. Never leave a group without agreed adult coverage. An older child may choose an easier drawing task; a younger child may tackle a deeper question orally.

# Choose by the next useful action

Offer one or two choices within a table. A child can point or dictate while an adult records. Reading speed, handwriting, and number of finished pages do not decide access.

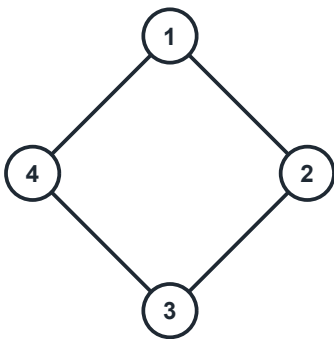
| Student pages / problems | What children do; useful entry knowledge                                                                                 |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------|
| 1-2 / P1-4               | Make, undo, and share puzzles. Match a picture; change both endpoints. No independent reading or arithmetic needed.      |
| 3-4 / P5-8               | Walk one light around a ring or tree; compare starting with one ON versus all OFF. Follow connected roads.               |
| 5-6 / P9-10              | Match many targets on larger rings and a grid. Use a legal pair move reliably; count only if helpful.                    |
| 7-8 / P11-14             | Try separated islands, draw a bridge, invent a board. A brief impossible attempt followed by a successful change.        |
| 9-12 / P15-18            | Record a move list; try pictured targets; inspect what one move does. Recognize labels 1-4; count to four or pair lamps. |
| 13-16 / P19-22           | Compare attempts, follow paths, save different pictures, explain the collection. Sort pictures; notice repeated states.  |
| 17-19 / P23-25           | Try a five-ring, find different lists, cancel duplicate presses. Track a short list; count to five.                      |
| 20-22 / P26-27           | Compare chosen roads with the roads left out. Tell a picture of selected roads from a picture of lit lamps.              |
| 23-25 / P28-29           | Make forced yes/no decisions; replay and compare results. Finish one lamp before attending to the next.                  |
| 26-28 / P30-32           | Rule out a target; invent short solutions; compare a six-ring. Follow the two decision branches and discuss why.         |
| 29-30 / P33-36           | Draw closed shapes, then split a square with one wall. Trace boundaries; adult can draw or count corners.                |
| 31-32 / P37-38           | Try two walls; compare three-wall maps. Find each room and mark it once.                                                 |
| 33-34 / P39-42           | Mark neighboring rooms, invent a town, make a shared town. Identify rooms sharing a piece of wall.                       |
| 35-38 / P43-46           | Separate components, add a bridge, cancel shared path edges. Follow lettered routes; count targets within each piece.    |
| 39-42 / P47-50           | Invent components; solve a tree by cuts. Judge even/odd groups by pairing; track the two sides of a cut.                 |

**Useful jumps:** p. 1 → p. 3 for a child who wants movement; p. 1 → p. 9 for a collector; p. 1 → p. 17 for someone already making reliable lists. Keep pp. 15-16 together: P22 uses the saved collection from P21. Start drawing at p. 29 or 30 at any time. A mathematician can introduce p. 35 after component questions arise. P26 and P28 each occupy two pages.

# Make, change, undo

Idea: a legal move changes both ends of one road. Repeating the move undoes it. Before any explanation, let children change the lamps themselves.

**Say:** “An empty lamp is OFF. A dot is ON. Choose a road. Change BOTH ends: add a dot where it was empty; erase a dot where it was ON. Leave the other lamps alone.” With counters: white is OFF and dark is ON. Use one method consistently at each board.



## P1: demonstrate and undo

Begin all OFF. Use road 1-2, then use it again: both lamps return OFF. Next use 1-2 and then 2-3: one endpoint turns OFF and another turns ON. Ask a child to replay. Point to both ends before changing either. Finish the whole move before choosing another road.

## P2: three pictured targets

ON 1, 2: **1-2**.

ON 1, 3: **1-2; 2-3**.

ON 1, 2, 3, 4: **1-2; 3-4**.

Reset all OFF between targets. These numbers belong to the adult key: 1 top, 2 right, 3 bottom, 4 left. “1-2; 2-3” means two roads in succession, not a list of lit lamps.

## P3: a one-move partner puzzle

The maker starts all OFF and makes one move while the partner watches. The partner undoes it; swap. A hint can simply point to the original road. Showing and replaying the move counts as a solution.

## P4: two moves, then three if wanted

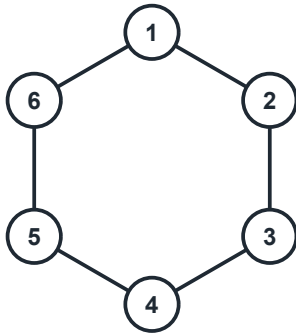
The maker starts all OFF and makes two moves. The partner tries to return all OFF. Reversing the original moves always works; a shorter solution may also work. Two uses of the same road leave the already-solved picture: welcome that discovery. Try three moves only when the pair wants more.

## Trio roles and a clear stopping point

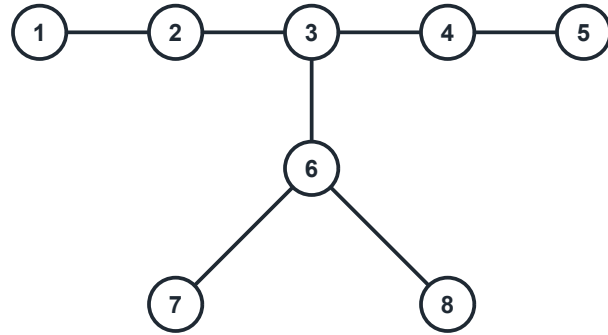
One child chooses the road; one changes one end; one changes the other. Rotate roles after each move. For partner tasks, the adult can partner with the third child. After two successful independent moves, offer a new page without requiring every target or a written list. If erasing is dominating, use the one-light walk (p. 3) or room drawing (p. 30).

# A light travels; a path leaves ends

Set up the starting picture by hand. This is different from reaching that picture through legal moves. The printed working boards are blank so children can erase every dot.



Six-ring: lamp 1 at the top



Tree: lamp 1 at the far left

## P5: reach a destination with exactly one light

Start with only 1 ON. To reach 3: **1-2; 2-3**. To reach 4: **1-2; 2-3; 3-4**. Reset for each target. Going the other way also works. Choose a road touching the lit lamp; erase that dot and add one at the other end. A finger can trace the route before the child starts.

## P6: tour the ring

Use **1-2; 2-3; 3-4; 4-5; 5-6; 6-1**. The light visits every lamp and returns home. Detours are allowed; no move list is required. Ask a partner to choose a different destination or a direction.

## P7: choose a branch

Reset with only 1 ON before each trip.

To 5: **1-2; 2-3; 3-4; 4-5**.

To 7: **1-2; 2-3; 3-6; 6-7**.

To 8: **1-2; 2-3; 3-6; 6-8**.

Backtracking is legal. Stop after one destination if that was enough.

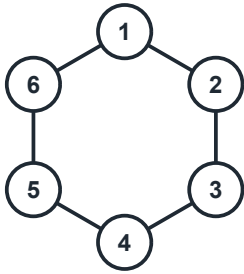
## P8: use the same route from a different start

**Reset all OFF.** To leave 1 and 8 ON, use **1-2; 2-3; 3-6; 6-8**. The starting lamp stays ON while the other light travels. Middle lamps change twice and end OFF. Compare this with the last trip in P7. "Watch the lamps behind you" is a useful hint.

**Optional depth:** Act out the same route with one light at its beginning, then with all lamps OFF. The difference is visible. Let the child explain by pointing; general parity language can wait.

# Bigger boards and invented puzzles

Use these pages when children want more lamps or room to invent. None is a gate to the collection, optimization, or drawing pages.



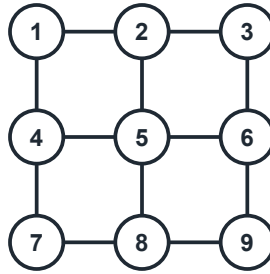
## P9: ring targets

ON 1, 2: **1-2**.

ON 1, 4: **1-2; 2-3; 3-4**.

ON 1, 2, 4, 5: **1-2; 4-5**.

ON 1, 2, 3, 4, 5, 6: **1-2; 3-4; 5-6**.



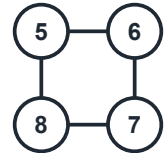
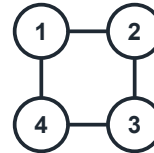
## P10: grid targets

ON 1, 5: **1-2; 2-5**.

ON 2, 8: **2-5; 5-8**.

ON 1, 3, 7, 9: **1-2; 2-3; 7-8; 8-9**.

ON 1, 3, 4, 6, 7, 9: **1-2; 2-3; 4-7; 6-9**.



## Help without taking over

Reset all OFF for each picture. Suggest a route between two desired lamps. A path leaves only its endpoints ON; try another path for another pair. On the grid, the top row is 1, 2, 3; the middle is 4, 5, 6; the bottom is 7, 8, 9. Longer working lists are welcome. The lists above are examples, not answers children must copy.

## P11: a short island mystery

Start with only lamp 1 ON on the left; try to leave only lamp 5 ON on the right. It is impossible on the printed map. The light has no road between islands. Even allowing extra lights cannot fix this: each move changes two lamps on one island, so the left island cannot change its odd count to zero. Keep the unsuccessful search brief; move to P12 after a few useful attempts.

## P12: change the map

Draw bridge **2-5**. Reset with only 1 ON. Use **1-2; 2-5** to carry the light to 5. Other bridges also work with suitable routes. Ask what changed; completing the new route is enough.

## P13-14: make a board and share it

**P13:** Draw 4-8 lamps connected by roads, with no crossings except at lamps. From all OFF make two moves, then let a partner undo the result. Reversing the moves always succeeds. **P14:** Add a lamp and a road connecting it to the old map; reset and make another puzzle. If a crossing is ambiguous, add a lamp there or redraw. An adult can draw while the child chooses.

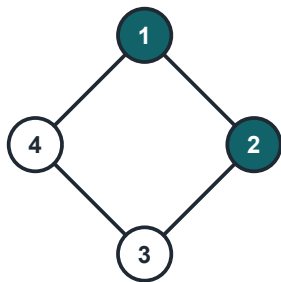
# Pictures first; a record that helps

Idea: compare reachable and apparently troublesome pictures, then examine what one move can change.

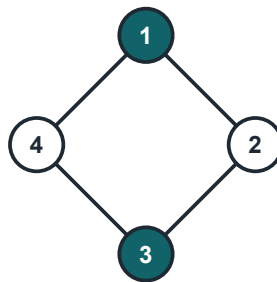
Readiness: follow labels 1-4 and count or pair up to four lamps.

## P15: introduce the move list through a replay

Start all OFF, use **1-2**, then **2-3**. The pictures are first {1,2} and then {1,3}. Show that “12, 23” records roads used, not the target lamps. Replay the two presses in reverse to return all OFF. For the child’s own list, accept any correctly recorded sequence that a partner can replay. The adult can write the numbers.



After road 1-2



Then road 2-3

## P16: four successful targets

{1,2}: 1-2. {1,3}: 1-2; 2-3.

{2,4}: 1-2; 1-4. {1,2,3,4}: 1-2; 3-4.

Reset all OFF each time. Any working list counts. A longer list can expose undoing or a different route; do not demand a shortest solution yet.

## P17: two successes and two obstructions

Targets {1} and {1,2,3} cannot be reached from all OFF. Targets {2,3} and {1,4} use 2-3 and 1-4. “Not yet” accurately records that the child has not found a list; it is not a proof of impossibility. Mix these attempts with successful ones and stop a fruitless search before it becomes frustrating.

## P18: three local changes

Set up each printed start **by hand**, including the one-ON picture; setup need not be reachable from all OFF. Flip road 1-2 once. At its endpoints: **0 ON** → **2 ON**, **2 ON** → **0 ON**, or **1 ON** → **1 ON**. Other lamps stay as they were. Children can point to the two affected lamps or pair the ON lamps instead of naming odd and even.

**Decision to continue:** If children are curious why some targets resist every attempt, use pp. 13-16. If recording is tiring, return to a bigger board or partner puzzle; the mathematical exploration still counts.

## Eight pictures, and why these eight

The recording grid gives twelve spaces so the page does not reveal the answer. Save the all-OFF picture deliberately; a blank unused space is different.

### P19: revisit the eight target attempts

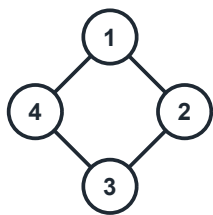
The pictured targets from P16-17 have ON counts **2, 2, 2, 4; 1, 2, 3, 2**. The six even targets can be made; the two odd targets cannot. Children mark their own progress and retry, rather than copying a finished classification. Ask them to compare the counts of pictures they made.

### P20: two routes leave two ends

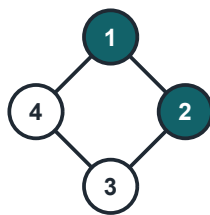
**1-2; 2-3; 3-4** leaves {1,4}. **1-4; 4-3** leaves {1,3}; 4-3 is the same road as 3-4. The middle lamps change twice; the two ends change once. Replay slowly and point to those lamps. This is a construction tool for the next page.

### P21: collect different pictures

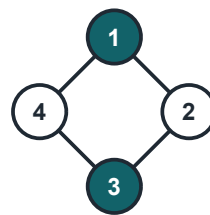
Exactly eight pictures are reachable: none ON; any of the six pairs; all four ON. Check “saved” for each recorded picture, including all OFF. A different move list that gives the same picture is not a new state. Help compare drawings without making counting or handwriting a gate.



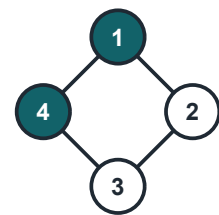
all OFF



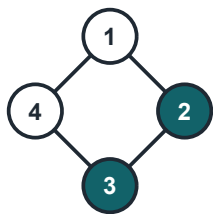
1, 2



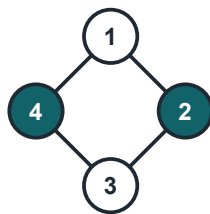
1, 3



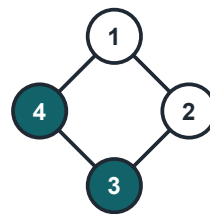
1, 4



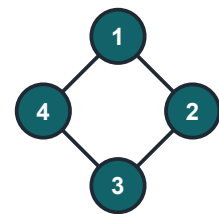
2, 3



2, 4



3, 4



1, 2, 3, 4

### P22: organize, then replay a construction

Counts are **0, 2, or 4**; the complete collection has eight pictures. A child can select any saved target and find a list for a partner to replay. For all OFF, no moves works; pressing one road twice also works. Invite the completeness question only after the collection has a purpose.

**Why complete?** Every move changes the ON count by +2, -2, or 0, so an odd count cannot arise from zero. All six pairs are made by paths; all four by roads 1-2 and 3-4. This both rules out every missing state and constructs every listed state. Pairing lamps or acting out the cases is a valid child-level explanation.

# Find two ways, then compare them

Children may start here after demonstrating the rule. Readiness: track short move lists, distinguish roads from lamps, and count presses. Writing can be shared with an adult.

## P23: a five-ring and undoing

Start all OFF. Press 1-2, then 1-2 again: return all OFF. Choose any two moves and undo them by reversing the order. Here and below road 5-1 is also written 51. Ring labels begin at the top and run clockwise.

| P24 target | Two lists, with number of presses              |
|------------|------------------------------------------------|
| {1,2}      | 1-2 (1); or 2-3, 3-4, 4-5, 5-1 (4). Minimum 1. |
| {1,3}      | 1-2, 2-3 (2); or 5-1, 4-5, 3-4 (3). Minimum 2. |
| {1,2,3,4}  | 1-2, 3-4 (2); or 2-3, 4-5, 5-1 (3). Minimum 2. |

## P24: let the experiment precede the lower bound

Reset before each list. Two different lists can tie (for example, 12,23 and 23,12); accept “same,” then invite a shorter or longer alternative. The singleton-edge target needs at least one press because the start is different. One press lights adjacent lamps, so cannot make {1,3}; it lights only two lamps, so cannot make the four-lamp target. These facts prove the shown two-press solutions shortest.

## P25: order and repeated presses

Lists 12,23, 23,12, and 12,23,12,12 all leave {1,3}. Removing two copies of the same road preserves the result. Each lamp only cares whether it was changed an odd or even number of times. Let children test before offering this explanation.

| P26 selected roads | Complement / common target / counts |
|--------------------|-------------------------------------|
| 12                 | 23,34,45,51 / {1,2} / 1 and 4       |
| 12,23              | 34,45,51 / {1,3} / 2 and 3          |
| 12,34              | 23,45,51 / {1,2,3,4} / 2 and 3      |
| 12,23,34           | 45,51 / {1,4} / 3 and 2             |

**P26 spans student pp. 20-21.** Thick roads mean “press once”; dashed roads mean “do not press.” These are not pictures of lit lamps. Reset before each half. Both complementary sets give the same target because pressing every ring road changes each lamp twice.

## Finish one lamp at a time

The decision method investigates all reduced solutions: each road used zero or one times. Lists with repeated roads are infinite in number; pairs of repeats can be cancelled.

### P27: invent a complementary pair

Choose one to three roads on the left; mark exactly the omitted roads on the right. Use every chosen road once, starting all OFF for each trial. Both give the same target, and the counts add to five. A working example is left **12,34**, right **23,45,51**, target {1,2,3,4}.

**Demonstrate the decision:** First fix whether 51 is used. Then decide 12 to leave lamp 1 matching its target. Decide 23 to finish lamp 2; then 34; then 45. Never revisit a finished lamp. Check lamp 5 last. Replay the completed list on the working board.

| Problem / target | 51  | 12  | 23  | 34  | 45  | Check / count |
|------------------|-----|-----|-----|-----|-----|---------------|
| P28 / {1,3}      | no  | yes | yes | no  | no  | works / 2     |
| P28 / {1,3}      | yes | no  | no  | yes | yes | works / 3     |
| P29 / {1,2,3,4}  | no  | yes | no  | yes | no  | works / 2     |
| P29 / {1,2,3,4}  | yes | no  | yes | no  | yes | works / 3     |
| P30 / {1}        | no  | yes | yes | yes | yes | fails / 4     |
| P30 / {1}        | yes | no  | no  | no  | no  | fails / 1     |

### P28: carry both branches through

On student p. 23, do not use 51; the first forced decision is to use 12. On p. 24, first use 51; the next forced decision is not to use 12. Both branches reach the target. Count the initial 51 when it is used. The order of the final chosen roads does not affect the final picture.

### P29: make four lights

On student p. 25, the two lists are **12,34** and **51,23,45**, with lengths two and three. The shorter list is shortest: one press cannot light four lamps. Ask the child to circle the shorter trial after replaying both.

### P30: check the last lamp honestly

On student p. 26, both decision branches leave {1,5}, so lamp 5 is wrong. The two branches exhaust all reduced possibilities; cancelling repeats shows no longer list can evade that failure. Pair parity gives a second explanation: one ON cannot arise from all OFF. End with the reason for failure, rather than more random trials.

# How short can every solution be?

Reserve deeper questions for children who want to explain the patterns. A child may instead construct and compare examples without completing a general proof.

## P31: invent targets and compare complementary lists

Any new target with two or four ON lamps on the five-ring works. One sample set of three targets is:

**{1,4}**: 51,45 (2) or 12,23,34 (3).

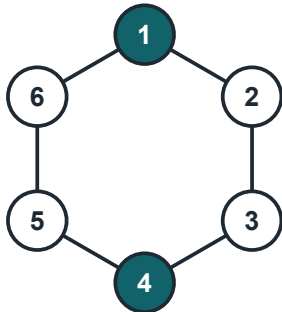
**{2,5}**: 12,51 (2) or 23,34,45 (3).

**{2,3,4,5}**: 23,45 (2) or 12,34,51 (3).

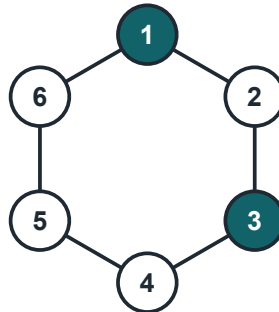
The second list must use every road omitted by the first; merely reordering the same roads does not test the complementary pattern.

## The five-ring has a two-press bound

Every reachable target has exactly two reduced solutions. Once 51 is fixed, successive lamps force the other four decisions, so there are at most two. Complementing any solution gives another one, because using every edge changes each lamp twice. Their sizes add to five: one is at most two. Cancelling pairs of repeated roads never makes a list longer, so the smaller reduced solution is globally shortest. No reachable target needs more than two presses.



P32: opposite lamps 1 and 4



P32: lamps 1 and 3

## P32: six lamps change the comparison

**{1,4}**: 12,23,34 or 61,56,45; both have three presses.

**{1,3}**: 12,23 or 61,56,45,34; lengths two and four.

The same forced-choice argument gives exactly these two reduced solutions, so no shorter list exists. The two complementary reduced solutions can tie on a six-ring; they cannot tie on the five-ring. More generally a ring with  $n$  edges gives complementary lengths  $k$  and  $n-k$ .

## Prompts that preserve the discovery

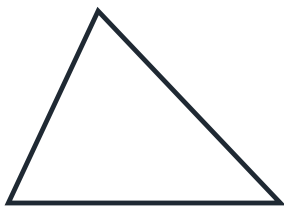
“What does using all the roads do?” “After deciding this road, is there any choice at the next lamp?” “Could a repeated road help make a shortest list?” Invite the child to demonstrate with a drawing or list. Keep the general argument as a discussion after enough examples, not a written requirement.

## Closed shapes and one-wall houses

A change of theme with the same paper or tablets. No lamp prerequisite. Trace, compare, and invent; adults may draw or record for children.

### P33-34: corners on a dot grid

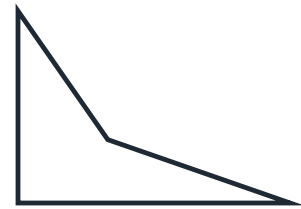
**P33:** Join three non-collinear dots with three straight sides to make a triangle; make another. **P34:** Join four dots in a closed four-sided shape without crossed sides; make another. Squares and rectangles count. A four-corner shape may have a dent. Count direction changes while tracing, not every grid dot along a straight side.



3 corners



4 corners



4 corners, a dent

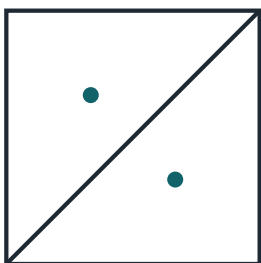
**Launch a house:** “Here is one room. Draw one straight wall from the outside edge to the outside edge. Put one dot inside every room.” Trace a room boundary together. The wall goes through the inside, has no gap, and does not follow the outside edge.

### P35: make four different one-wall houses

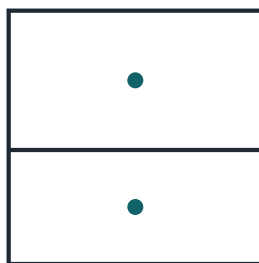
Every full wall makes **two rooms**. Vary its direction or endpoints. Different-looking examples are useful even when their underlying structure is the same. Dot each room before counting aloud. Neatness is not the mathematical problem; an adult can straighten a child’s chosen wall.

### P36: change the shapes of the rooms

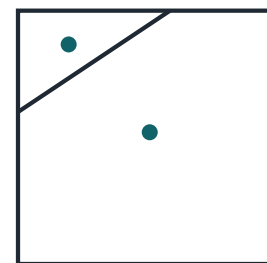
A diagonal between opposite corners makes **two triangles**. A wall joining opposite sides away from corners makes **two four-corner rooms**. Children may make or find these examples. Optional follow-up: a wall cutting off one corner makes **a triangle and a five-corner room**. Trace each boundary and tap each corner.



3 + 3 corners



4 + 4 corners



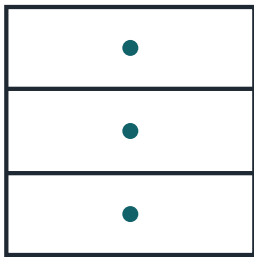
3 + 5 corners

# Room counts worth comparing

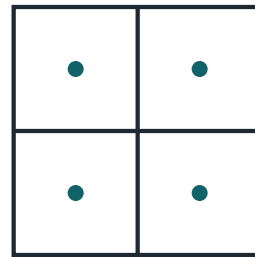
Each picture needs its own concrete counting action: one dot in each room, then touch each dot. The adult can supply a number after the child finds the rooms.

## P37: two distinct full walls

Try four drawings. Two walls make **three rooms** if they do not cross inside the square, or **four rooms** if they do. Meeting only on the boundary is not an interior crossing. Tracing an old wall does not add a new wall. Ask for a different room count only after the child has counted a picture successfully.



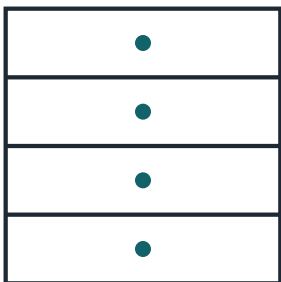
3 rooms: no crossing



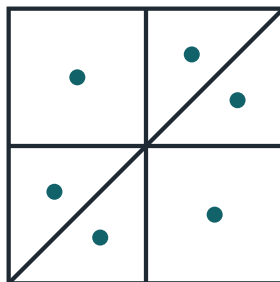
4 rooms: one crossing

## P38: count three printed towns, then invent

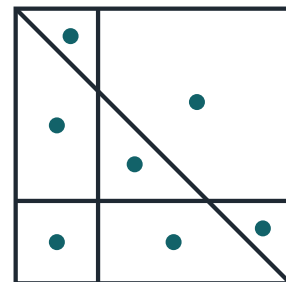
The printed maps have **4, 6, and 7 rooms**, in that order. In the middle, all three walls meet at the same point. The last has three distinct crossings and a central triangular room; mark that room too. For the blank town, any three distinct full walls are valid. No maximum or complete list is required.



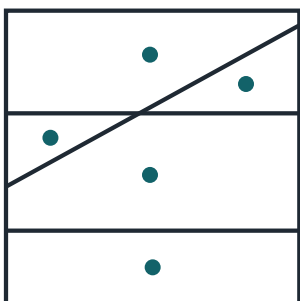
4 rooms



6 rooms



7 rooms



Optional: 5 rooms

## If someone asks for a missing number

Three full walls can also make **five rooms**: the example crosses the slanted wall with only the upper horizontal wall. Three distinct full walls can therefore make 4, 5, 6, or 7 rooms.

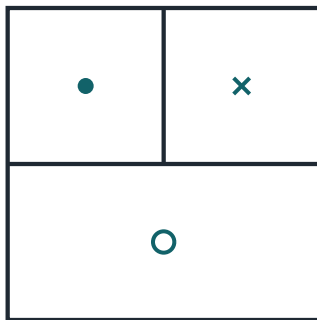
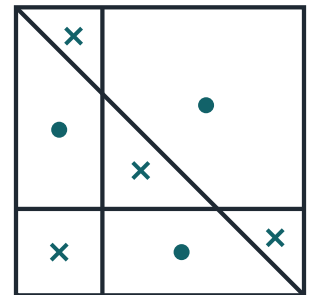
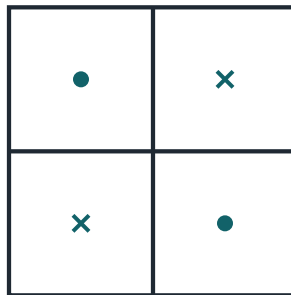
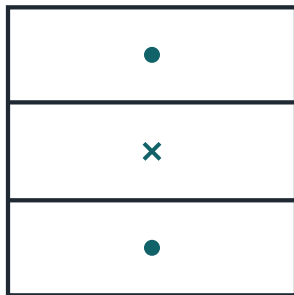
Follow a new wall from edge to edge. Each piece between distinct interior crossing points cuts an old room in two. With three walls there can be at most  $1 + 1 + 2 + 3 = 7$  rooms. A repeated crossing point counts once along the new wall.

# Neighbors and a town to share

Offer room marking once children can find the rooms. These are choices for any table; a shared whiteboard town is also a good regrouping activity.

## P39: a dot or an X in every room

Rooms sharing a piece of wall must have different marks. Touching only at a corner is allowed. These are valid two-mark solutions for the three printed maps. Swapping every dot and X gives another. Demonstrate one neighboring pair first.



## P40: a T-shaped surprise

This map has a wall that stops at another wall. Every room touches both of the others along a wall. If the top rooms are dot and X, neither mark works below. **Two marks cannot work; adding a circle works.** Keep the attempt short, then let the child finish with a third symbol.

Three mutual neighbors

## P41: invent a partner's town

Draw one, two, or three distinct full straight walls. A partner dots and counts the rooms, then the children swap. Counts can range from two through seven. Trace questionable boundaries together. Dot/X marking can be an extra oral invitation after counting; it is not needed to finish this task.

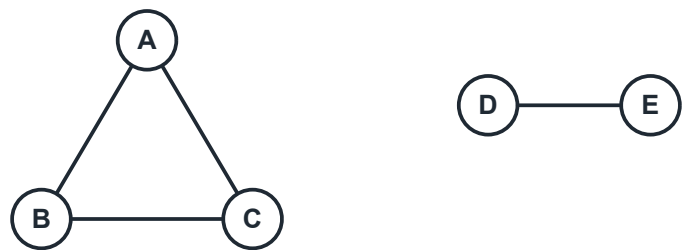
## P42: a giant shared town

Draw one large square on paper or a whiteboard. Take **three turns**, adding one new full wall per turn, then stop. Do not trace an old wall or steer toward a maximum. With more than three children, share drawing and marking roles. Each child can dot an undotted room; count together. Optional next round: try dot/X marks.

**Why two marks work for full walls:** Choose one side of each wall. Give each room a dot or X according to whether it lies on an even or odd number of chosen sides. Crossing a wall changes exactly one side choice, so adjacent rooms differ. The T has an interior endpoint and three mutually adjacent rooms, so this argument does not apply.

# When the map falls into pieces

Reserve route: compare connected pieces before introducing tree cuts. Children should track lettered roads and count or pair targets within a component.



| P43 ON target | Sample moves / outcome                              |
|---------------|-----------------------------------------------------|
| {A,C}         | AC                                                  |
| {D,E}         | DE                                                  |
| {A,D}         | Impossible: one target in each component.           |
| {A,C,D,E}     | AC; DE                                              |
| {A,B,C,D}     | Impossible: three in the triangle, one in the edge. |
| {A,B,D,E}     | AB; DE                                              |

## P44: count inside each piece

For the first six targets in order, counts in triangle ABC / edge DE are **2/0**, **0/2**, **1/1**, **2/2**, **3/1**, **2/2**. The added all-OFF target has **0/0** (no presses); {B,C} has **2/0** (press BC). Exactly the rows with an even count in each component work. Two targets in total are not enough: {A,D} separates them into different components. Let the child point to the unmatched target in each piece.

## P45: add the bridge CD

Now {A,D} uses **AC; CD**. Target {A,B,C,D} uses **AB; CD**. Both previously failed. Adding the bridge makes one component, so every even target is reachable. Reset all OFF for each target.

## P46: overlapping paths cancel

The paths A-C-D and B-C-D give **AC; CD; BC; CD**. The repeated CD can be removed twice, leaving **AC; BC**. Both lists light exactly A and B. Count a shared road twice before cancelling; taking the ordinary union of path edges would leave the wrong endpoints.

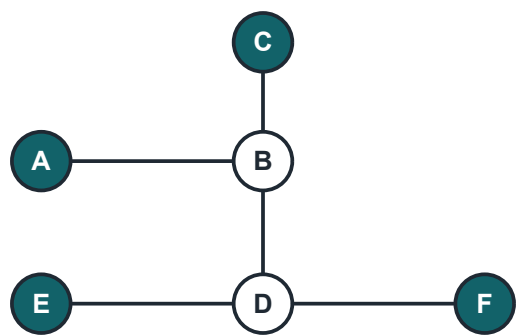
**Complete component rule:** Starting all OFF, each move changes two lamps in a single component, preserving evenness there. Conversely, pair target lamps within each component and use a path for each pair. Interior lamps cancel and the paired endpoints remain. Shared path edges can be pressed twice or cancelled. Isolated lamps are components too and cannot be turned ON.

# Road choices on a tree

P47 offers invention before the tree investigation. Cut reasoning is an optional deeper conversation; give it enough time after successful component examples.

## P47: invent two components

Draw two separate connected pieces, each with at least three lamps. A possible target has an even number in each piece; one example selects two lamps in each. An impossible target can select one lamp in each: even overall, but odd in both pieces. Pair and connect desired lamps by paths to verify the possible example. Preserve the child's invented map for later use.



## P48: two targets on this tree

**{A,C}**: AB; BC. **{A,C,E,F}**: AB; BC; DE; DF.  
The picture above shows the second target. Both lists leave B and D OFF. Repeated moves may yield longer lists, but after cancelling pairs there is exactly one solution on a tree.

| P49 road removed | Named side | Targets there | Use this road? |
|------------------|------------|---------------|----------------|
| AB               | A          | 1             | yes            |
| BC               | C          | 1             | yes            |
| BD               | A,B,C      | 2             | no             |
| DE               | E          | 1             | yes            |
| DF               | F          | 1             | yes            |

## P50: predict a new even target

For any new even target, remove each road mentally and count target lamps on one side. Use the road exactly when that count is odd. For example, target **{A,E}** needs **AB; BD; DE**. Target **{B,D}** needs only **BD**. Replay to check the prediction. Either side of a cut gives the same odd/even answer when the whole target is even.

**Why unique and shortest:** In a tree, the removed road is the only road across that cut. Moves within one side change two lamps there; only the cut road changes its parity. Its use is therefore forced by the target count. This determines every road. Path pairing guarantees a solution exists. Each required road must appear an odd number of times in any list, so using each required road once is shortest.

# Keep the evidence and the next questions

This shared edition is a response to the organizer's printed-page feedback and approval of a shared collection. Its success in this classroom has not yet been observed.

## Source findings that informed the design

**Math Circle by the Bay**, preface, pp. viii-x; especially p. x: the authors describe using the same material across ages at different pace and depth, with younger groups sometimes proceeding equally quickly. The shared collection is our adaptation; the book does not establish this staffing plan.

**Natasha Rozhkovskaya**, *Math Circles for Elementary School Students*, **Lesson 3**, “At the lesson,” item 1: children tried and explained before a table was introduced; reserve problems served a mixed-experience group. **Lesson 7**, items 1-2: missed legal moves affected a game investigation; open polygon construction was engaging. Drawing polygons instead of using K'nex is our adaptation. **Lesson 8**, item 2: writing and coloring pace split the class. Our adult scribing and readiness-based page choices respond to that report.

## Local continuity, with a limit on the claim

The project README and Week 1 classroom review record the need for concrete attempts, time with materials, and explicit next actions. Lowell year-1 **Handout 11**, **Problem 11.3**, has invented shapes split and recombined for a partner; **Handout 9**, **Problem 9.1**, toggles doors. The pair-road rule, component test, and optimization here ask different questions. A handout's presence in the collection does not show which children encountered it.

## Further mathematical connections

Toggling adds by symmetric difference: using something twice cancels. Edge subsets with prescribed odd-degree vertices are T-joins. The cycle and tree arguments in this guide prove our instances directly; they do not claim to prove the general matching algorithm. Adult references: Edmonds and Johnson, *Matching, Euler tours and the Chinese postman* (1973), doi:10.1007/BF01580113; Cornuéjols, *Combinatorial Optimization: Packing and Covering*, Chapter 2, pp. 27-29. For matrix context, MIT 18.06SC, “Graphs, Networks, Incidence Matrices”; the binary adaptation used here is ours.

## Record actual use, not the whole available collection

Log the date, shared edition ID, exact problem numbers and targets tried, and invented maps worth keeping. Note whether children made a legal move unaided, replayed a record, found every room, explained a failure, or asked to repeat something. Separate observations (“erasing took most of the turn”) from interpretations (“the recording method may be too slow”). Mark conclusions as tried, conjectured, checked in cases, proved, or supplied.

## Returning children and the next revision

Keep unused pages for later. Revisit the theme with a different graph, question, representation, or proof depth; do not simply reissue every target. Preserve drawings and memorable explanations. Expand a useful idea that needed more attempts; simplify a record that needed repeated adult rescue. Finishing fewer pages while making and explaining a discovery is a successful outcome.

# Choose a substantial problem

September 30, 2026. **Unpiloted.** This revision applies the current Week 2 catalogs and AGENTS.md: short shared rules, several contrasting examples under one question, and methods left for children to choose. The pages are a choice library for an hour, not a completion assignment.

## Prepare for ten children and three adults

Current roster: KK1 / 3333 / 445. Keep one adult at each table: the parent with the three youngest, the other mathematician with the four third graders, and the organizer with the three oldest. Offer pages across these groups when useful. Print three K first pages, four middle first pages, and three upper first pages: **ten starting sheets**. Bring a master of each continuation and this guide for the adults; children can continue on blank paper or existing whiteboards. No classroom printer is assumed. Paper, pencils/erasers, whiteboards and counters suffice. Optional hand-drawn cards make the action tangible: two sets of three shape cards per K child (18 cards total); a set of A–D for each middle child (16); A–E for each upper child (15), with F–I in reserve. Repeated symbols can be drawn on spare scraps. Keep an input row visible while building a separate output row.

## Brief whole-group launch, then two work periods

**0–4 minutes:** let children make messages with the materials. **4–8:** show the shape key on K page 1. Send circle–triangle–circle through it, making triangle–square–triangle below it. One turn changes each old symbol once; it does not repeatedly change a newly written symbol. Let a child make the next turn. Show that an empty row before an arrow asks for the input. On a small letter-key example, show that a column pairs one input with one output. Then distribute pages; do not demonstrate a solving strategy. **8–28:** one chosen investigation. **28–33:** stand, stretch, reset. **33–50:** continue or offer another question. **50–55:** share an attempt or discovery. **55–60:** tidy. These are adjustable planning intervals. An absorbing problem can take both work periods.

## Prerequisites and useful first offers

**K, Problems 1–2:** match shapes; adult reads; no arithmetic. Problem 3 adds counting to three and tracking a whole message. Problems 4–5 concern recoverable information.

**Middle, 1:** A–D are labels, not words; coordinate equal symbols and a one-to-one rule. Problems 2–5 add counting to six and comparing constructions.

**Upper, 1–2:** track five letters, count to six, and eventually justify that all cases are covered. Problems 3–4 are an alternative direction in composition.

**Extra:** common returns through 20 and organizing a completeness argument; no algebra notation or factorial prerequisite. Offer after a child understands whole-key return. A message, a useful impossible clue, or the five-versus-six-turn comparison can be a complete session. Allow roughly 8–15 minutes per ordinary investigation, and 15–30 or more for a maximum or completeness proof. Adults can read and scribe. Finishing one page is not a gate to another.

# Shapes: messages and information

## K 1–2: send, recover, invent

Read each arrow as one turn. The six missing rows in Problem 1, from top to bottom, are:

| Row | Missing message          |
|-----|--------------------------|
| 1   | triangle–square–triangle |
| 2   | circle–circle–square     |
| 3   | square–triangle–circle   |
| 4   | circle–square–circle     |
| 5   | triangle–triangle–square |
| 6   | square–circle–triangle   |

The first three rows ask for outputs, the last three for inputs. Repeated symbols must be treated consistently. Problem 2 sustains the same mathematics through partner invention: send a message, pass only the output, and recover the original using the printed key. Rotate sender, receiver, and checker at a three-child table. There is no required message length. **If needed:** cover all but one column of a message and ask which shape produces the visible output. Ask a child to replay a disputed turn. **Later question:** can the receiver ever have two possible originals with this key? Each output has exactly one input, so no. This explanation can use pointing alone.

## K 3: whose return are we counting?

The three starts are square; circle–square–circle; circle–triangle–square. Under P, each returns first at turn **3**. Under Q, their returns are **1, 2, 2**. A return is after at least one turn; an unchanged square returns in one, not zero. Allow children to choose cards, drawings, or a list. Do not require six copied trace tables. If a child has lost the starting message, keep a separate copy. If counting one shape hides the rest, ask whether the *whole* message matches its start. To introduce loops later, replay circle’s three turns, placing arrows between the visited shapes; only then show that a counter can follow the same route. Let the child try that record from triangle and square. **Extension:** make a message that returns after one turn under Q, and one that takes two. Exactly the nonempty messages made entirely of squares have first return one. The others take two. Under P every nonempty message takes three, because no shape is fixed by one or two turns.

## K 4: recoverable keys

A key must have one output for each input. It lets a partner *always* recover the message precisely when the three outputs are all different. The six possible output rows, in input order circle/triangle/square, are the six arrangements of these shapes. Identity is allowed; it returns in one turn. Three keys swap a pair and fix the third (return two); two cycle all three (return three). Children need not find all six. If they want completeness, choose the circle’s output in three ways, then the triangle’s in two, leaving one square output. If a child reuses an output, test two different one-shape messages. This reveals the issue through action instead of a premature prohibition. Optional extension: find every recoverable key and compare return times.

## K 5: a key that loses information

Both circle and triangle become square; square becomes circle. The four inputs for square–square are circle–circle, circle–triangle, triangle–circle, and triangle–triangle. Each position has two choices, independently; no input square is possible. If needed, fix the first shape while changing the second. Extension: three output squares have eight possible inputs. A receiver cannot select one original without additional information.

# Four letters: hidden keys and returns

## Middle 1: every key, or none

Output rows below are in input order A,B,C,D. Each clue is a separate puzzle.

| Clue | All possible output rows | Reason                                        |
|------|--------------------------|-----------------------------------------------|
| a    | CDAB, CDBA               | A and B fixed by the clue; two outputs remain |
| b    | BCDA                     | A,B,C fixed; one output remains               |
| c    | none                     | The two A's would have different outputs      |
| d    | none                     | A and B would share output C                  |
| e    | ABCD, ABDC               | A,B fixed; C,D can stay or swap               |
| f    | DBAC                     | A,B,C fixed; one output remains               |

The blank table holds a candidate; extra keys can go below or on a whiteboard. Clue c breaks consistency: identical inputs need identical outputs. Clue d breaks reversibility: distinct inputs need distinct outputs. If needed, ask which letter causes a conflict. Extension: invent clues with one, two, or no keys.

## Middle 2-3: incomplete messages can hide the key

| Starting message | P | Q | R |
|------------------|---|---|---|
| D                | 1 | 2 | 4 |
| ABBA             | 3 | 2 | 4 |
| ABCD             | 3 | 2 | 4 |

P has the loop A–B–C–A and fixed D. Q swaps A/B and C/D. R has one four-letter loop. Under P the full rows are ABCD, BCAD, CABD, ABCD. A fixed D alone does not certify that the whole key has returned. For Problem 3, choose output rows **BCAD** and **BCDA**. Both send ABBA to BCCB, but ABCD returns in three and four turns respectively. The missing letters C and D leave room for different keys. If needed, ask which key entries the message ABBA actually tests. Extension: can two distinct keys agree on a message containing every letter? No: that message tests every input.

## Middle 4-5: construction and completeness

The possible first returns for ABCD are **1,2,3,4**. Example output rows: ABCD, BACD, BCAD, BCDA. A message containing every letter returns exactly when the whole key does. Four letters have these five possible loop-size lists:

| Sizes  | 4 | 3+1 | 2+2 | 2+1+1 | 1+1+1+1 |
|--------|---|-----|-----|-------|---------|
| Return | 4 | 3   | 2   | 2     | 1       |

No list gives six. Adding E allows A–B–C–A and D–E–D, output **BCAED**; the first simultaneous return is six. Merely testing many keys does not prove that four letters cannot work. Use the loop argument below and exhaust the five size lists if the child wants a proof. Do not supply these lists before the construction attempts.

## Why loops and common returns settle the question

Follow a symbol until one repeats. The first repeat must be the start: a later first repeat would give a symbol two predecessors, contradicting reversibility. Remove that loop and continue with unused symbols. This splits the key into disjoint loops, including fixed letters as one-loops. A loop returns at multiples of its length; the whole key returns at the first common multiple of all lengths. If needed, move counters around a three-loop and a two-loop and mark home arrivals. “Least common multiple” names this result; it is not an entry requirement.

# Five letters and key order

## Upper 1–2: a maximum without a prescribed method

The printed R,S,T,U keys first return after **5,6,4,2** turns. S is slowest among these. Its full trace is ABCDE, BCAED, CABDE, ABCED, BCADE, CABED, ABCDE. R is one five-loop; S has separate loops of sizes three and two; T has sizes four and one; U has sizes two, two, and one. For Problem 2, possible returns are **1,2,3,4,5,6**. Witness output rows are ABCDE, BACDE, BCADE, BCDAE, BCDEA, BCAED respectively. The complete loop-size classification is:

| Sizes  | 5 | 4+1 | 3+2 | 3+1+1 | 2+2+1 | 2+1+1+1 | 1+1+1+1+1 |
|--------|---|-----|-----|-------|-------|---------|-----------|
| Return | 5 | 4   | 6   | 3     | 2     | 2       | 1         |

To prove completeness, use the loop argument on page 3. A largest loop of five or four leaves only one split; three leaves two or one+one; two leaves two+one or three ones; one forces all singletons. Therefore six is the maximum, and no time is missing from the listed possibilities. **Hints, only if needed:** follow one letter; ask what happens to unused letters; ask how to organize the keys already found. If checking all keys becomes copying, draw just their loops. A child may find another valid classification. Blank workspace deliberately replaces the former largest-loop table.

## Upper 3: overlapping keys need a real test

P swaps A/B; Q swaps B/C; L swaps D/E. V and W are opposite three-cycles on A,B,C, fixing D,E.

| Pair | First then second | Second then first |
|------|-------------------|-------------------|
| P,Q  | CABDE             | BCADE             |
| P,L  | BACED             | BACED             |
| V,W  | ABCDE             | ABCDE             |

P and Q do not commute: A goes through B to C in one order, and through A to B in the other. P and L commute because they affect disjoint sets of letters. V and W *overlap* but commute: either order undoes the three-cycle. Thus “overlap always changes the answer” is false. An exact rule is: reversing the order leaves every message unchanged *between the two orders* precisely when the two routes give the same final letter for *each* input letter. Checking ABCDE tests every input. Disjoint action is sufficient, not necessary. Children may reach a sufficient rule before an exact one; distinguish their scope. There is no requirement to classify all commuting permutations. **If needed:** keep the intermediate message visible; test one disputed letter. **Extension:** invent another overlapping pair that does commute (a key and its inverse, or two powers of the same key), and one that does not.

## Upper 4: the combined machine and its inverse

P followed by Q sends ABCDE to CABDE. Repeating this whole two-key turn gives BCADE and then ABCDE, so its first return is **three machine turns**, or six individual key applications. The combined key has the loop A–C–B–A and fixes D,E. To undo one P-then-Q turn, use **Q then P**. Each swap undoes itself, and the inverse reverses their order. From CABDE, Q gives BACDE and P restores ABCDE. Every input letter is tested, so the same inverse works for any message. Optional extension: a general two-key machine is undone by the second key’s inverse, then the first key’s inverse.

# Long returns and complete bounds

## Extra 1: construct before optimizing

With A–H, suitable loop sizes for returns 4,6,8,12,15 are respectively **4+4, 6+2, 8, 4+3+1, 5+3**. Put successive letters around each loop to obtain actual keys. These are examples, not unique answers. Their output rows can be BCDAFGHE, BCDEFAHG, BCDEFGHA, BCDAFGEH, and BCDEAGHF. Check the 12-turn example as A–B–C–D–A, E–F–G–E, H fixed. If needed, first build one loop and then two independent loops. Two four-loops return together after four turns, not sixteen; the return is a common multiple, not automatically the product. Let children compare attempts before naming LCM. Extension: find other loop-size lists giving the same return.

## Extra 2: records for eight and nine letters

Eight letters achieve **15** with 5+3; nine achieve **20** with 5+4. The proof needs both a construction and a bound. Following the loop theorem, classify by the largest loop. These are adult solutions, not a template to hand out at the start.

| Letters | Largest loop | Greatest return in that case |
|---------|--------------|------------------------------|
| 8       | 8, 7, 6      | 8, 7, 6                      |
|         | 5            | 15 (5+3)                     |
|         | 4            | 12 (4+3+1)                   |
|         | at most 3    | 6                            |
| 9       | 9, 8, 7, 6   | 9, 8, 14, 6                  |
|         | 5            | 20 (5+4)                     |
|         | 4            | 12 (4+3+2)                   |
|         | at most 3    | 6                            |

For eight letters with largest loop five, the remaining three split as 3,2+1,1+1+1, giving returns 15,10,5. With largest four, the leftover four split as 4,3+1,2+2,2+1+1,1+1+1+1, giving 4,12,4,4,4. Larger loops leave only one or two letters, giving the displayed bounds. If every loop has size at most three, every length divides six, so the return is at most six. For nine letters with largest five, the remaining four give returns 20,15,10,10,5. With largest four, all remaining loops have lengths 1–4, so their common return divides twelve; 4+3+2 attains twelve. With largest seven, the leftover two give at most fourteen; with largest six, all leftover lengths divide six. The cases cover every key, proving both optima.

## Extra 3: an extra letter need not improve the record

For sizes 1 through 9 the maxima are **1,2,3,4,6,6,12,15,20**. The five-to-six-letter plateau disproves strict growth. For six letters, largest loops 6,5,4,3,2,1 give case maxima 6,5,4,6,2,1: for largest three the leftover three can supply a two-loop, attaining six. For seven letters, largest loops 7,6,5,4, at most 3 give maxima 7,6,10,12,6; 4+3 attains twelve. Sizes up to five follow by the earlier partition arguments (sizes one to three have maxima one, two, three). The record can *never decrease*: keep an optimal old key and make the new letter fixed. Its return time is unchanged. This proves monotonicity for every size, beyond the nine checked values. Do not describe nine examples alone as proof of a universal statement.

# Sources, continuity, and next evidence

## What changed, and why

The organizer approved the concise September 30 Week 2 catalogs: short prompts can support sustained construction, impossibility, optimality, and completeness problems. Week 3 now groups contrasting cases under substantial questions. Shared rules appear once; letter tables specify keys or leave room for candidates. Blank work areas allow children to choose how to record. Tracing every turn, drawing loops, sorting by largest loop, and marking common multiples are available adult supports rather than printed procedures. The extra packet retains proof as its explicit task. Earlier pages also retain central impossibility and completeness questions. Concision removes repeated instructions, not mathematical depth. These changes are design proposals; no classroom result for this revision is claimed.

## Source passages reviewed locally

**Natasha Rozhkovskaya, *Math Circles for Elementary School Students*:** Lesson 2, Problems 2.2–2.8 (EPUB OEBPS/part0012.xhtml) supplies coded messages and repeated-letter clues. Its lesson report describes strong reasoning by children who were not yet fluent readers. Our short letter-label clues and arbitrary reversible keys adapt that entry; we are not reproducing the shift-code worksheet. Lesson 3, “At the lesson,” item 1 (part0013.xhtml), describes letting children attempt and explain before introducing a table. Lesson 8, item 2 (part0018.xhtml), reports writing and coloring pace separating groups. Our choice of optional adult records and generous unstructured workspace responds to these accounts; the sources do not establish that our particular pages will work. ***Math Circle by the Bay, preface, printed pp. viii–x (PDF pp. 9–11)***: substantial themes, manipulatives, clear statements, independent solving, reserve challenges, and different pace and depth across ages. Our exact prompts, timing, staffing, and five-minute-plus design aim are applications, not prescriptions quoted from the book. **Lowell year 1, Handout 8, Problem 8.1**: divisibility in sharing apples among possible guest counts provides continuity with common returns. The saved handout does not establish which children used it. This week changes the representation and question to reversible machines and maximum return time. Week 4 then restricts arbitrary keys to fixed shifts around a circle.

## Mathematical lineage

The existing design relates these tasks to permutations, disjoint cycles, inverses, and products (Judson, *Abstract Algebra: Theory and Applications*, Chapter 5, §5.1). The explicit loop proof in this guide is self-contained. Maximal permutation order is Landau’s function: the exact finite question in the extra packet. The prior design’s reference is Deléglise, Nicolas, Zimmermann, *Landau’s function for one million billions*, 2008 manuscript, §1.1–1.3. We retain that lineage; this revision makes no claim about current research status. The small values here are established by the displayed finite arguments and separately verified by enumeration.

## Record only what happened

Use edition IDs F03-K/M/U/X-v4 and the actual problem, key, and starting message in the use log. Record whether children changed each old symbol exactly once, distinguished message return from whole-key return, invented a useful record, found counterexamples, or needed repeated adult rescue. Save interesting invented keys. Mark a result as tried, conjectured, checked in cases, proved, or supplied. Keep observed actions separate from explanations and proposed fixes. Record what children wanted to continue. On a future return, use new alphabets, composition questions, or larger-size records instead of automatically repeating these same keys. Unused pages remain reserve material.

# Prepare the ring shifts

v3 revision, September 27, 2026: unpiloted. This applies Week 1's classroom guidance to new ring and shuffle work; no outcomes from teaching this revision are known.

## Materials and prerequisites

Print first pages for three K, three middle, and one upper child (seven starting sheets). Keep continuation masters ready to copy as needed, including the later ring mats. Leave the extra in reserve. Bring seven counters, pencils, scratch paper, and cards 0–7; cards 8–15 extend the reserve. A paper strip numbered 0–7 distinguishes card places from labels. No wheels or brads are required.

**K:** match four shapes, count hops to three with support. **Middle:** labels A–J, count to ten; read-aloud works. **Upper:** count to eighteen and compare routes; general gcd proof additionally requires division and adult-supported coprime divisibility. **Extra:** track positions and repeated doubling with remainders; exponent notation is optional.

## Handle, then launch: 0–15 minutes

Give 0–10 minutes to explore counters on rings. At 10–15 minutes, gather around the four-shape mat. Define the arrow direction by moving circle to triangle. Show a two-hop turn, counting only movements, and point to square as the landing; triangle was passed over. Reset and let a child replay it. Draw only the landing after the starting circle. Take one more two-hop turn and record its landing. Now distribute different starting tasks. The purpose is a shared action and record, not a gcd explanation.

## A flexible hour

**15–35:** K compares starts under one/two hops; middle uses the common shift and records two routes; upper explores +4 and +8. Adult A anchors K; organizer checks landing records with older children. Compare routes when useful. **35–40:** stand, stretch, reset. **40–55:** continue experiments or introduce one new record when useful. **55–60:** share and tidy. Page counts are reserves, not a completion schedule.

## Stops and later directions

**K:** stop after one-hop and two-hop comparisons; three hops and hidden-turn undo are later choices. **Middle:** stop after +2 routes from A,B,C,D. Marking the two sets helps explain a split already encountered; +3,+5 and inverse checks follow. The optional complete list of full-visit steps is an adult extension.

**Upper:** +4/+8 loops are a satisfying stop. Compare +3,+6,+5 before the summary table (7). Skip filling known entries if copying slows the child. Offer distance records only after physical routes; the strip/gcd proof is optional adult work. **Extra:** obtain several real rows and position tracks before the map, loops, or doubling rule. Stop at the eight-card return if that is enough.

## What to listen for

“Did you land there or pass over it?” “Which start is unused?” “Is this a card label or its place?” All shifts can be undone; a short orbit does not mean an undecodable key. Discuss why an argument covers every start only after children can show examples. Proofs and checked outputs follow.

# Exact routes and the gcd rule

## K-1

From circle, one hop visits triangle, square, star, circle (four turns); two hops alternate circle/square (two turns), never landing on triangle or star. A secret circle-to-square shift is two hops, so triangle becomes star, square becomes circle, and star becomes triangle. Two hops undo themselves. Three-hop turns follow circle, star, square, triangle, circle, visiting all four. On six positions, among steps 1 through 5, only 1 and 5 visit all six; 2 and 4 give two loops of length 3; 3 gives three loops of length 2.

## Middle

A to D is +3 on A-J. GDG decodes to **DAD**; BAG encodes to **EDJ**. B becomes E, so the additional B-to-F clue contradicts the fixed-shift rule. Repeated +2 gives A,C,E,G,I,A and B,D,F,H,J,B. Marking alternate letters explains why the two routes cannot meet: adding two preserves that marking. Repeated +3 gives A,D,G,J,C,F,I,B,E,H,A. The steps visiting all ten are **1,3,7,9**. Any shift is reversible: +2 is undone by -2, or +8. Skipping some letters along one repeated orbit is not information loss.

## Upper: key and complete classification

9 to 1 gives +4, which takes 2 to 6. Outputs 1,4,10 decode to **9,0,6**. Its four loops are (0,4,8), (1,5,9), (2,6,10), (3,7,11), each returning in three turns.

| Shift k         | 0  | 1  | 2 | 3 | 4 | 5  | 6 | 7  | 8 | 9 | 10 | 11 |
|-----------------|----|----|---|---|---|----|---|----|---|---|----|----|
| Return time     | 1  | 12 | 6 | 4 | 3 | 12 | 2 | 12 | 3 | 4 | 6  | 12 |
| Number of loops | 12 | 1  | 2 | 3 | 4 | 1  | 6 | 1  | 4 | 3 | 2  | 1  |

Steps 1,5,7,11 visit all twelve. The zero rule is the identity and has first *positive* return time 1, not zero. For eighteen positions, +6 returns after 3; +5 after 18.

## Why the formula works

For +8 on 12 places, return means  $3 \mid 2t$ . Draw three t-strips, totaling  $3t$ . If the first two strips total a multiple of 3, removing them leaves  $t$  a multiple of 3. Thus the first positive return is 3. This proves this case for every  $t$ . The optional general bridge is on page 4; do not treat a completed table as its proof.

# From turning to shuffling

## Upper: inverse and composition

On twelve places, +4 is undone by +8 (or -4). For every starting  $x$ , +4 followed by +7 gives  $x+11$  modulo 12; reversing the order gives the same because integer addition commutes. Every shift is bijective, even if its orbit from 0 is a proper subset. The visit-everywhere condition is  $\gcd(n, k) = 1$ , not a condition for reversibility. This distinction is central: a substitution permutation can have several cycles.

## Extra: eight cards, three shuffles

Rows, left to right with the leftmost card the top:

| Shuffle | Row             |
|---------|-----------------|
| 0       | 0 1 2 3 4 5 6 7 |
| 1       | 0 4 1 5 2 6 3 7 |
| 2       | 0 2 4 6 1 3 5 7 |
| 3       | 0 1 2 3 4 5 6 7 |

The old-to-new position map is **0,2,4,6,1,3,5,7**. Its cycles are fixed 0, fixed 7, (1,2,4), and (3,6,5). Thus the whole shuffle returns after 3 and not earlier. This is also a direct week 3 loop argument. For old position  $x$  in the first half (0–3), the new position is  $2x$ . In the second half (4–7), it is  $2(x - 4) + 1 = 2x - 7$ . For  $x$  below 7 these are exactly  $2x \bmod 7$ . Position 7 stays 7, separately; it must not be merged with residue 0. Repetition gives  $2^t x \bmod 7$ . The powers 2,4,8 have remainders 2,4,1, so the first identity occurs at  $t=3$ . Tracking position 1 rules out earlier global returns, and remainder 1 fixes all positions 0–6.

## Extra predictions

Six cards use modulus 5. Powers 2,4,8,16 give remainders 2,4,3,1, so the order is **4**. Sixteen cards use modulus 15: 2,4,8,16 give 2,4,8,1, again order **4**. In general an out-shuffle on  $2m$  cards maps positions below  $2m - 1$  by doubling modulo  $2m - 1$ , fixes the final position, and has the multiplicative order of 2 as return time (the two-card identity is a trivial edge case). More cards can therefore have the same or even a smaller return time.

## A useful additional contrast

Random shuffling asks about a distribution over possible orders. This perfect shuffle is one precisely defined permutation repeated deterministically. Its return time says nothing by itself about how mixed a random shuffle is. Do not introduce chance unless it becomes a separate, explicitly changed rule.

# Mathematical lineage and reuse

## Optional adult bridge: coprime divisibility

If positive  $a, b$  are coprime and  $a \mid bt$ , compare base strips  $a, b$  with scaled strips  $at, bt$ . Repeatedly subtract the smaller base strip from the larger, doing the same to the scaled strips. Both scaled lengths stay multiples of  $a$ . The positive base sum decreases, so the process ends with equal lengths  $c$ . Common divisors are unchanged: a divisor of  $x, y$  also divides  $x-y, y$ , and conversely by adding. Thus  $c=1$ , since the original lengths were coprime. The scaled process ends at  $t$ , which is therefore a multiple of  $a$ . Conversely,  $a \mid t$  plainly implies  $a \mid bt$ .

Try 5,3 first:  $5-3=2$ , then  $3-2=1$ . On scaled strips this leaves  $3t - (5t - 3t) = t$ . Equivalently,  $2 \cdot 3t - 5t = t$ ; if  $5 \mid 3t$ , the remainder is a multiple of 5. Ask the child to explain each removal. This is additional readiness, not assumed knowledge of prime factors.

For  $k>0$ , set  $d = \gcd(n, k)$ ,  $n = da$ ,  $k = db$ . Return means  $a \mid bt$ ; the lemma gives  $a \mid t$ , hence first return  $n/d$ . Every start has this length, so there are  $d$  loops. For  $k=0$ , every point is fixed. If only the 3-and-2 case was explained, record that proof and label the general rule conjectured or given as a theorem. Full strip prompts and the Week 9 application: `plans/coprime-divisibility-scaffold.md`.

## Sources and boundaries

**Judson, *Abstract Algebra: Theory and Applications***, Ch. 4, §4.1, Theorem 4.13, pp. 48–49, gives the cyclic order formula: textbook. Our rings are its additive version. **Diaconis, Graham, Kantor (1983)**, *The mathematics of perfect shuffles*, §2, Lemma 1, p. 177, proves the out-shuffle order via doubling: original paper. The extra develops this small return-time case and position rule, not the paper's shuffle-group classification. The plan has full research references. Rozhkovskaya, Lesson 2, problems 2.3–2.7, supplies the shift-cipher entry. *Math Circle by the Bay*, preface viii–x, supports manipulatives, changing pace, and reserves; Rozhkovskaya Lesson 8 records copying slowing the activity. Our rings, proof scaffolds, and hour are adaptations. Lowell Handout 6, problem 6.7, and Handout 7, problem 7.8, use a different card-ordering solitaire. Ask about recognition; do not assume this shuffle was taught.

## Use history and checks

Record actual steps, ring sizes, and extensions used as F04-K/M/U/X-v3. Keep 26-letter Caesar messages as a later alternative, not extra arithmetic for early finishers. Future years can compare prime/composite ring sizes, affine maps, in-shuffles, or perfect-shuffle groups; avoid replaying the same small cases. `verify.py` checks all 12 shifts, the 10-letter routes, message examples, the 18-ring predictions, shuffle rows, position formulas, and first return times for 6, 8, 16 cards. The facilitator proofs establish why the patterns hold.

# v3 checks and observations

## Added ring experiments

**K 1–2:** from circle land triangle,square,star,circle; from triangle land square,star,circle,triangle. **3–4:** circle and square share the two-hop orbit; triangle and star share the other. Changing from circle to square never reaches triangle. **5–7:** the three-hop, hidden shift, and inverse results are on page 2. Read and scribe the shape records; writing is not a gate.

**Middle 1:** +2 takes A to C and I to A. **2:** +3 takes B,I,J to E,B,C. **3–4:** messages and incompatible clue are on page 2. **5–7:** C follows C,E,G,I,A,C, the A-route; D follows D,F,H,J,B,D, the B-route. The two markings are preserved by every +2 turn. **8:** full +3 route is on page 2. **9:** +5 gives A,F,A and B,G,B, length 2. **10:** +2 then +8 gives A,C,A; B,D,B; I,A,I; J,B,J. Each starting letter returns despite the shorter orbits.

## Upper: new comparisons

**1:** +4 sends 0,9,2,8 to 4,1,6,0. **4:** +8 traverses each +4 loop in reverse. **5:** +3 gives three loops of length 4, +6 six loops of length 2, +5 one loop of length 12. **7:** full table is on page 2. **8:** eighteen-place +6 route is 0,6,12,0; +5 is 0,5,10,15,2,7,12,17,4,9,14,1,6,11,16,3,8,13,0. **9:** distances at turns 1–4 are 4,8,12,16 and 8,16,24,32; first whole laps are turn 3 in both. **10:** +4 then +8 returns all starts, with intermediate positions 4,5,9,3. **11:** both orders total +11, sending 0,2,9 to 11,1,8. **12:** in general shifts +a,+b compose as +(a+b), wrapping modulo 12.

## Extra: position scaffolding

**1–6:** page 3 gives all eight-card rows, map, loops, doubling proof, and fixed last position. Position traces from starts 1,3,7 are 1,2,4,1; 3,6,5,3; and always 7. **7:** six-card rows are 012345,031425,043215,024135,012345. **8:** sixteen-card loops are (0),(15),(1,2,4,8),(3,6,12,9),(5,10),(7,14,13,11). The full return is 4. **9:** 6 and 16 cards both take 4; 8 cards take only 3. Ask for a proof after these checks: position 1 rules out earlier returns, and the same remainder-1 multiplier fixes all other positions below the last.

## Evidence for the next revision

Record actual step sizes/decks attempted, mistaken starting-point counts, skipped-versus-landed confusion, success using a route record, and card-label/position confusion. Note exact adult rescue and what children wanted more time for. Distinguish observations from hypotheses, and cases checked from arguments proved. The extra and general theorem may remain unused; this is not incomplete work. v3 is unpiloted.

# A city is a mathematical object

**Question:** How much information forces one arrangement? Children progress from a complete list to a uniqueness proof to a lower bound on clues. These are questions about permutations, Latin squares, and determining sets.

## Preparation and prerequisites

Use snap cubes: six per youngest child, eighteen per middle child, forty for the upper child: **112 cubes**, plus a few for the launch. Prebuild towers if time is short. Numeral slips can record a second solution after the physical attempt. Start with page 1 for each child (seven student sheets). Keep continuation masters and this guide ready; copy the next page when the group needs it. Build on the tabletop or quick scrap-paper grids with one-inch cells. The smaller printed grids record solutions; they are not sized to hold every brand of cube.

K-1: compare heights/count to 3; no independent reading; recognize different orders. Middle: count to 3 and follow row/column rules; adult may read; distinguish an example from a complete list. Upper: count to 4; track constraints, cases, and uniqueness; no formal algebra. Extra: understand a 4-by-4 Latin square, coordinate labels, and a pigeonhole argument.

## Launch

Let towers be handled first. Show 2,1,3 from the left. Ask which towers a tiny person sees; touch each visible tower. Walk to the other end. Only after this introduce the separate city rule: every row and column uses each height once. A clue counts visible towers, not the first tower's height or the number of cubes.

## A shared hour with seven children and two adults

0-10: explore materials freely. 10-15: brief launch and rules. 15-35: main investigation. 35-40: movement/reset; preserve work. 40-55: continue or choose an optional proof. 55-60: share and tidy. This flexible default is our proposal, not a source prescription.

One adult stays with K,K,1. The organizer alternates middle and upper, leaving a concrete next attempt and returning for an explanation. In games keep two opposing sides and rotate a checker. Proof continuations need adult attention; pages are not a completion quota.

**K-1 stop:** Build all six skylines and describe their views. **Optional proof:** organize by first tower to show completeness; explain the impossible one-at-each-end view.

**Middle stop:** Build both top-clue cities and select one with the second clue. **Optional proof:** certify the list, then swap unseen rows/columns to rule out every one-clue puzzle.

**Upper stop:** Explain the complete five-clue case table. **Optional proof:** check all five deletion certificates and prove the three-clue alternative forces the same city. The computer's minimum claim is separate.

# Complete lists and minimum clues

## K-1: six skylines, not six unrelated puzzles

The complete list (left count, right count) is: 123 (3,1), 132 (2,2), 213 (2,1), 231 (2,2), 312 (1,2), 321 (1,3). Fix the first tower: each of three choices leaves two orders. This proves completeness without factorial notation. One visible tower from both ends is impossible with distinct heights: the tallest would have to occupy both ends. Both 132 and 231 show two from each end.

The two 2-by-2 cities are 12/21 and 21/12. The top-row left clue 2 chooses the former. If row/column rules are too much, keep investigating rows; a complete skyline list is already substantial mathematics.

## Middle: exactly two cities, then exactly one

Top clue 3 forces first column 1,2,3. Top row is either 123 or 132. Column exclusions force the respective completions:

|   |   |   |
|---|---|---|
| 1 | 2 | 3 |
| 2 | 3 | 1 |
| 3 | 1 | 2 |

|   |   |   |
|---|---|---|
| 1 | 3 | 2 |
| 2 | 1 | 3 |
| 3 | 2 | 1 |

The right clue 2 on row 2 keeps the first. Deleting that clue leaves both. Deleting only the top clue permits 123/231/312 and 312/231/123 (swap rows 1 and 3).

**A genuine minimum proof:** No single visibility clue can determine an entire 3-by-3 Latin square. If it sees a row, swap the two other rows; if it sees a column, swap the two other columns. The clue stays true and the square changes. Two clues suffice in our example, so the minimum over all uniquely solvable 3-by-3 visibility puzzles is two.

## Hints and evidence

K: fix the first tower and exchange the other two. Middle: point down the first column, then try both top rows. Ask what rules out another answer. Accept a complete physical case split; long written prose is unnecessary. If children only solve the city, ask for one clue deletion and an alternative, saving the minimum proof for conversation.

# A human certificate for five clues

## Upper: the unique solution

Both the opening and the five-clue puzzle have rows **1234/2341/3412/4123**. Here is a small exhaustive argument for five clues. Bottom row starts with 4 and has right visibility 2, so 3 must be last. Only 4123 and 4213 remain. Keep both: the bottom clues alone do not rule either out; row 3 and row 2 do that next.

Row 3 has two views of 2. Its six possible rows are **1423, 2143, 2413, 3142, 3241, 3412**. Cross out candidates conflicting with the bottom row. The surviving pairs and possible row 2s are:

| Bottom row | Row 3 | Row 2 with right view 2 |
|------------|-------|-------------------------|
| 4123       | 3241  | none                    |
| 4123       | 3412  | 2341                    |
| 4213       | 3142  | none                    |

Check the final column by listing row permutations avoiding the two entries already in each column, then testing the right view. The top row is forced by missing column heights. This proves uniqueness without trusting a solver.

## Deleting clues: witnesses

Deleting L3, L4, R2, R3, R4 gives **2, 2, 6, 12, 20** solutions respectively. L3 means the left clue on row 3. One alternative per deletion proves necessity:

| Delete | Alternative city, top row first |
|--------|---------------------------------|
| L3     | 1234 / 3412 / 2341 / 4123       |
| L4     | 1234 / 4123 / 3412 / 2341       |
| R2     | 1324 / 2431 / 3142 / 4213       |
| R3     | 1324 / 3142 / 2431 / 4213       |
| R4     | 1234 / 2143 / 3412 / 4321       |

On optional upper page 4, check each certificate is Latin, keeps all four retained clues, and differs from the target. Restoring the clue before the next deletion keeps the five arguments separate. The five-clue set is **minimal**, but it is **not minimum**: left clues 4,3,2 on rows 1,2,3 uniquely determine this same target. First row is 1234. The rows with left view 3 are 1243,1324,1342,2134,2314,2341; all but 2341 repeat a column entry. With rows 1 and 2 fixed, the possible row 3s are 3412 and 4123; only 3412 has left view 2. The last row is forced.

The script exhaustively verifies that three is the minimum *for this target city among subsets of its 16 perimeter clues*. That computation is not a proof that every 4-by-4 city needs three. For children, the concrete three-clue alternative already disproves the claim that minimal means minimum.

# Trades, bounds, and research

## Extra: a lower bound with an escape move

Each bold block has form  $ab/ba$ . Exchanging  $a, b$  inside it preserves the multiset in each affected row and column. With at most three entry clues, one of four disjoint blocks has no clue. Switch it: a distinct Latin square keeps every given entry. Thus at least four *entry* clues are necessary for this square.

Hitting these four blocks is not sufficient. Clues at  $(1,1), (1,3), (3,1), (3,3)$  leave rows 2 and 4 unclued; swapping those rows keeps all four clues. Any uniquely determining set must meet *every* trade. Conversely, the changed cells in any alternative completion form a trade; hitting every such trade rules out alternatives.

**Keep clue types separate:** Previous pages use outside visibility clues; the extra uses fixed entries. Bounds do not transfer automatically.

## Where the mathematics goes

Undergraduate combinatorics studies permutations, Latin squares, constraint satisfaction, and hitting sets. Joy Morris's *Combinatorics*, chapter on Latin squares develops Latin-square structure; our visibility questions add a different constraint system.

Hatami and Qian, *Teaching dimension, VC dimension and critical sets for Latin squares* (2016), §1 and Theorem 1.2, study sets of *entries* determining a square and prove a quadratic lower bound for sufficiently large orders. Our trade argument is a small instance of the question, not their asymptotic theorem. The paper states a sharper conjecture; no current open-status claim is needed here.

## Teaching sources and returning children

JRMF *Skyscrapers Teaching Guide*, PDF pp. 1–4: physical viewpoints, city rules, and reasoning; pp. 13–14: blank boards. Our clue instances and investigations extend that format. *Math Circle by the Bay*, printed pp. ix–x: manipulatives, extra challenges, and themes at different depths. Lowell Handout 2.1–2.2 seating constraints are related prior material, not evidence children have solved these cities.

Use F05-K-v3/M-v3/U-v3/X-v3. Prepared, not recorded as taught. Save trade classification, Latin transversals, and larger skyline recurrences for later years. Rebuild with `build.sh`; `plans/verify-week-05.py` checks the finite claims independently of the explanations.

# Using the revised sequence

## Classroom evidence and this new proposal

Revision v3, September 27, 2026: **unpiloted**. Week 1 feedback reported too little work before new ideas, unclear next actions, and simultaneous adult demand. It did not report a Week 5 trial. These revisions apply that guidance by giving each viewpoint several attempts and each later representation a job. Preserve children's actual records; do not equate an unfinished packet with a failed session.

## Launch and handoff

After handling the cubes, gather everyone around towers 2,1,3. Touch the towers visible from each end; record 2,1,3 left to right and write its two view counts. Then change to 1,3,2 and let a child check. Distribute first pages only after children can make and record that action. Demonstrate the row/column city rule locally when a group reaches a grid; the K group can spend the whole session on rows.

K pages 1–2 provide repeated views and a complete list. Problem 1 has views (2,1), Problem 2 has (2,2). Problem 3 accepts 123 for three; 312 or 321 for one; 132,213,231 for two. Problem 6 has no solution; preserve two unsuccessful attempts before discussing the tallest-at-both-ends argument. Page 3's city rule is a separate reserve.

Middle page 1's four rows have view pairs (3,1), (2,1), (2,2), (1,3). Any Latin city and any whole-row swap meet Problem 2. Pages 2–3 stage the top clue, a second clue, and a deletion, with separate grids. Stop after the two top-clue cities if comparison still needs time. Ask the completeness and unseen-row/column-swap proofs orally only after the children have built the alternatives.

Upper page 1's row views are (2,2), (2,2), (1,2), (1,2). Page 2 isolates possible rows before page 3 organizes them into cases. The complete lists and table on this guide's page 3 answer Problems 4–7. Pages 4–5 are optional clue investigations. The covered-clue views for Problem 9 are respectively **3,3,3,3,4**; each violates the restored target clue while retaining the other four. A smaller construction and a lower bound are different claims. Preserve the minimal/minimum discussion in conversation after the checked deletion records and three-clue construction.

## Extra and practical pacing

Extra page 1 tests four individual switches before entry clues appear on page 2. For any three circled cells, at least one of four blocks is untouched; any such switch works. For Problem 4, use the original city and **1234/4321/3412/2143**, obtained by swapping rows 2 and 4. Each has the four fixed entries. Invite the four-block lower-bound and every-trade arguments from page 4 of this guide after these concrete checks.

Allow 15–35 minutes for first attempts, a 35–40 movement break, and 40–55 for the current idea or one selected continuation. Keep 55–60 for sharing. Do not start two adult-dependent proof discussions together. Offer pointing, building, or dictation for every recording task.

## Record after use

Record exact problem numbers attempted, viewpoint/row-rule errors, any instructions needing repeated adult rescue, what children chose to continue, and explanations actually given. Keep observations separate from hypotheses and proposed revisions. Source support: *Math Circle by the Bay*, pp. ix–x; Rozhkovskaya, Lessons 3,7,8, “At the lesson.” These support flexible pacing and representation after attempts; the exact sequence is our adaptation.

# Useful questions; provably few tests

**Question:** Can we design a method that works for every secret, and prove no shorter method can always work? This is decision-tree complexity with exact-position feedback.

## Prepare and check the rules

Use two kinds of counters (mark R/B or distinct symbols), paper position strips, and three folded-paper screens. Use 12 counters of each kind for K and middle, and 16 of each kind for upper so six candidate rows and a test can remain visible. Keep all records. A parent may write dictated guesses and scores. First score visible secret RBR against RRR (2), BRB (0), and RBR (3). Repeated colors are allowed. Give only the **total correct positions**; no wrong-place-color feedback.

Start with page 1 for each child (seven student sheets). Keep the remaining pages as continuation masters, including any earlier records children will compare. Print the extra only when selected. K: no reading; match positions and count 0–2; eliminate impossible examples. Middle: counts to 3, optional R/B reading; compare one changed feature and complete cases. Upper: counts to 4; worst-case reasoning, symmetries, and branching. Extra: 0/1 notation and small equations; variables can be replaced by counters.

## Count the same thing throughout

A **test** is a scored query. Once exactly one secret remains, naming it is free in this investigation. Original game rules require actually submitting the secret, potentially one extra turn. Our exact optima for 2, 3, 4 positions are **2, 3, 4 tests to identify**, not claims of those same turn limits for winning submission. Stop early if a score already identifies the secret.

## A shared hour with seven children and two adults

0–10: explore materials freely. 10–15: brief launch and rules. 15–35: main investigation. 35–40: movement/reset; preserve work. 40–55: continue or choose an optional proof. 55–60: share and tidy. This flexible default is our proposal, not a source prescription.

One adult stays with K,K,1. The organizer alternates middle and upper, leaving a concrete next attempt and returning for an explanation. In games keep two opposing sides and rotate a checker. Proof continuations need adult attention; pages are not a completion quota.

**K-1 stop:** Build a complete two-test method. **Optional proof:** for any first test, build two secrets scoring one, so one test cannot always work.

**Middle stop:** Explain why the three-test method works. **Optional proof:** combine the one-B triple obstruction with color renaming to cover every first test. Without renaming, claim only the RRR-first obstruction.

**Upper stop:** Explain why four tests always suffice. **Optional proof:** defend the triple and four-candidate branches separately, then use symmetry to prove optimality. Choose one older group's proof conversation at a time.

# Two and three places: complete proofs

## K-1: elimination becomes a decision tree

RR scores 0 only for BB, 2 only for RR, and 1 for RB or BR. In the last branch test RB: scores 2 or 0 distinguish them. So two tests suffice. One cannot always work: for every first test, two secrets match in exactly one position. Build those counterexamples.

## Middle: a three-test upper bound

Let baseline RRR score  $s$ , the number of Rs in the secret. Flipping one test position R to B changes its contribution from 1 to 0 if the secret there is R, or 0 to 1 if it is B. The new score is  $s - 1$  or  $s + 1$ , never unchanged. Tests BRR and RBR reveal positions 1 and 2. Their number of Rs subtracted from  $s$  determines position 3. This procedure covers all eight secrets.

## Middle: why two tests cannot guarantee success

After RRR scores 2, candidates are BRR, RBR, RRB. The complete second-test table is:

| Test | BRR | RBR | RRB |
|------|-----|-----|-----|
| RRR  | 2   | 2   | 2   |
| RRB  | 1   | 1   | 3   |
| RBR  | 1   | 3   | 1   |
| RBB  | 0   | 2   | 2   |
| BRR  | 3   | 1   | 1   |
| BRB  | 2   | 0   | 2   |
| BBR  | 2   | 2   | 0   |
| BBB  | 1   | 1   | 1   |

Every row repeats a score; some answer leaves two candidates. Conceptually, the three secrets differ only in the position of one B. A test only reveals whether that B falls inside its B positions, giving at most two outcomes. For the numbered symmetry step, physically turn BRB into RRR by swapping R/B at places 1 and 3 in both secrets and queries. An equal pair stays equal and an unequal pair stays unequal. Every score is preserved. Do this at the B positions of any first query; it becomes RRR, so the lower bound covers all first queries and all adaptive methods.

## Hints

Make candidate rows tangible. The table is optional once a child explains the one-B structure and two possible scores; further entries add no proof. Test a troublesome pair first. Ask what score a test would earn for each candidate. A test need not itself be a possible secret to reveal useful information.

# Four tests: upper and lower bounds

## Upper bound

RRRR and three separate single-position flips determine places 1–3 by score changes; the baseline R count determines place 4. Four tests suffice. Generally this uses  $n$  tests to identify an  $n$ -bit secret, at most  $n + 1$  turns if winning submission is required.

## A difficult answer that stays consistent

By independent color renaming, take the first test to be RRRR. Answer 2, leaving six two-B secrets. Classify the second test by its B count. Position permutations and complementing a test (replacing score  $s$  by  $4 - s$ ) reduce all cases:

**0 or 4 Bs:** no information; six remain.

**1 B (BRRR):** scores 3 or 1 each leave three candidates. Take score 3: BBRR, BRBR, BRRB. Ask: *Which position is fixed? What is the only feature that changes?* They share the first B; the other B occupies one of three places. Any last test has a fixed first-position contribution and only two possible scores on the remaining one-B triple. It cannot separate all three. The score-1 branch has the same structure; 3-B tests are complementary.

**2 Bs (BBRR):** score 4 leaves BBRR, score 0 leaves RRBB, score 2 leaves BRBR, BRRB, RBBR, RBRB. Choose the four-candidate branch. For any last test with  $k$  Bs and a two-B secret, let  $j$  positions have B in both. The score is  $2 - k + 2j$ . There are at most three possible scores, so four candidates cannot all be separated.

Every second test has a consistent branch that needs more than one further test. Thus three tests cannot guarantee identification; four is optimal. The lower bound uses actual score structure, not merely five possible answers per test.

## Facilitating a hard proof

Let the child physically sort six code cards into score piles. Ask them to make the largest pile small, then play the stubborn setter. Upper pages 1–2 supply a satisfying stop. Pages 3–4 are an optional adult-supported lower-bound investigation; the final-query triple argument is different from the four-candidate counting argument. If full optimality is too much, obtain a proof of the four-test method and one failed three-test plan. Record that boundary honestly.

**Relationship to Week 3:** Here the hidden object is an ordered binary string, queries may adapt, and the goal is a worst-case measurement optimum. Score structure and adversarial lower bounds deepen earlier identification ideas.

# Five bits and a research landmark

## Extra: decode and prove

The example is **10110**, with scores 2,2,2,0. Let  $w$  count secret 1s and  $s_0, s_1, s_2, s_3$  be scores in printed order. Then  $w = 5 - s_0$ , and

$$x_4 + x_5 = \frac{s_1 - s_0 + 2}{2}, \quad x_3 + x_5 = \frac{s_2 - s_0 + 2}{2}, \quad x_2 + x_5 = \frac{s_3 - s_0 + 2}{2}.$$

Each 0-to-1 change gains one match for a secret 1 and loses one for 0. If any pair-sum is 0 or 2, it fixes  $x_5$ , then  $x_2, x_3, x_4$ ;  $w$  fixes  $x_1$ . Otherwise all three sums are 1. If  $x_5 = 0$ , the middle bits are 111 and  $w$  is 3 or 4. If  $x_5 = 1$ , they are 000 and  $w$  is 1 or 2. The ranges do not overlap. So  $w$  fixes the case and  $x_1$ .

This proves four *preselected* tests distinguish all 32 secrets. The page claims sufficiency, not an independently proved five-place optimum. Matching scores encode binary-hypercube distances; distinguishing landmarks form a resolving set.

## Undergraduate and research connections

Decision trees, adversary arguments, Hamming distance, and information bounds belong to discrete mathematics and algorithm analysis. The small optima here have explicit upper and lower proofs.

Cáceres et al., *On the Metric Dimension of Cartesian Products of Graphs* (2005 preprint; 2007 publication), §2, PDF p. 3, gives exactly our four five-bit landmarks. §6, PDF p. 8, equates exact-position static Mastermind with resolving sets in Hamming graphs. This directly adapts a finite construction from a research paper, with our elementary decoding proof.

Knuth, *The Computer as Master Mind* (1976–77), gives a five-guess strategy for commercial four-place, six-color Mastermind; see Knuth's author page and strategy link. That game also has wrong-place-color feedback. Its result does not directly apply here. The shared idea is minimizing the worst surviving candidate set.

## Teaching lineage and future use

JRMF *Crack the Code*, PDF pp. 2–4 supplies binary exact-position play and the  $n + 1$ -turn challenge. The optimum/adversary tasks and extra deepen it. *Math Circle by the Bay*, pp. ix–x supports manipulatives, varied pace, and optional challenges.

Use F06-K-v3/M-v3/U-v3/X-v3; prepared, not taught. Save more colors, noise, and average-case analysis for later. `plans/verify-week-06.py` searches all adaptive query trees for small optima and checks all 32 extra signatures.

# Make the scores usable first

## Revision v3: unpiloted

The September 27 revision applies the Week 1 classroom guidance; no Week 6 classroom outcomes have been reported. We added repeated visible scoring, contrasting single-position changes, physical candidates, and separate recording space before trees and adversary arguments. These are design hypotheses to test, not claims of observed success.

## A concrete whole-group launch

After handling and copying counters, keep RBR visible. Build RRR alongside it and point to matches in positions 1 and 3. Say only “2” and record “RRR: 2.” Compare BRB (0) and RBR (3). Let a child check a test and dictate the log entry. Hide a secret only when this action is understood. Give first pages to groups; demonstrate branches locally after candidate filtering, never as the opening notation.

K page 1: against RB, tests RR,RB,BR,BB score **1,2,0,1**. Page 2: RR gives **2,1,1,0** against RR,RB,BR,BB. On remaining RB/BR, RR scores (1,1), RB scores (2,0), and BR scores (0,2). Page 3 introduces the tree to reuse those trials. It gives one test for RR/BB and two for RB/BR. The one-test lower bound remains an oral follow-up after testing the tree on all four secrets.

Middle page 1: visible RBR has scores 2,0,3. Page 2’s rows BRR,RBR,RRB,BBB have score triples **(2,3,1), (2,1,3), (2,1,1), (0,1,1)**. Ask what a gain/loss says about the changed position before requesting the general method. A working three-test method is a satisfying stopping point. Page 3 uses a small selection of tests plus children’s own before the optional full-table argument. In Problem 6, swapping positions 1 and 3 sends RRB,BBB,BRR to BRR,RBR,RRB; every old and new score is 2. Checking examples motivates the symmetry proof on page 2 of this guide; examples alone do not prove that all first tests are covered.

Upper page 1: score pairs for RBRB,BRBR,RRRR,BBBB are **(2,1), (2,3), (4,3), (0,1)**. Upper page 2’s four-test records for RBRB,BRBR,RBBB,RRRR are **(2,1,3,1), (2,3,1,3), (1,0,2,2), (4,3,3,3)**. Keep each comparison anchored to RRRR. Naming is free throughout.

## Plan the adult conversations

Use 0–10 handling, 10–15 shared launch, 15–35 play/comparison, 35–40 movement, 40–55 current work or one continuation, 55–60 sharing. Starting-page print plan: seven sheets plus continuation masters. K may stop after consistent filtering; middle/upper may stop with a reliable method and its explanation. Return earlier score records alongside later pages. Do not require every table row if the child can already explain its structure.

Observe whether children score positions consistently, compare with the correct baseline, and use records to choose a next test. Record actual examples and claims established, not merely pages distributed. Distinguish observed difficulties, hypotheses about them, and proposed changes. The Week 1 review and Rozhkovskaya’s Lessons 3,7,8 motivate this order; the exact tasks are our adaptations.

# New tables and proof handoffs

## Upper pages 3–4: checked score records

In candidate order BBRR,BRBR,BRRB,RBBR,RBRB,RRBB, second-test rows are:

|      |   |   |   |   |   |   |
|------|---|---|---|---|---|---|
| BBBB | 3 | 3 | 3 | 1 | 1 | 1 |
| BRRR | 4 | 2 | 2 | 2 | 2 | 0 |
| BRRB | 3 | 3 | 1 | 3 | 1 | 1 |
| BBBR | 2 | 2 | 2 | 2 | 2 | 2 |

For the triple BBRR,BRBR,BRRB in Problem 7, final tests RRRR,RBRR,RBBR give **(2,2,2), (3,1,1), (2,2,0)**. For the quartet BRBR,BRRB,RBBR,RBRB in Problem 8, RRRR,BRRR,BBRR give **(2,2,2,2), (3,3,1,1), (2,2,2,2)**. Children's added tests vary; check by counting matches directly.

After these trials, offer the complete lower-bound conversation on page 3 of this guide. A finite collection of failed final tests is not enough. Establish separately why the moving-B triple allows at most two scores and why a two-B quartet allows at most three. Then cover every second test by B-count/position symmetry, and every first test by independent color renaming. Without the last step, report only an RRRR-first obstruction. Keep this distinction in the use log.

## Extra: three pages before the general decoding argument

Page 1 keeps all four probes visible through three examples and two actual games. Probe order is 00000, 00011, 00101, 01001. The provided secrets have records:

|       |   |   |   |   |
|-------|---|---|---|---|
| 00000 | 5 | 3 | 3 | 3 |
| 11111 | 0 | 2 | 2 | 2 |
| 10010 | 3 | 3 | 1 | 1 |

Page 2 first isolates two changed positions. Secret pairs 00,01,10,11 have 00/11 match counts (2,0),(1,1),(1,1),(0,2); changes are **-2,0,0,+2**. Thus score change reveals pair sums 0,1,1,2. The decoding example has total 1-count 3 and pair counts **1,1,0**; it gives 10110 as before. Keep physical counter pairs available before writing equations.

Page 3 gives the ambiguous pair-count case its own examples. If all three pair sums are 1, the four words are **01110,11110,00001,10001**, with total 1-counts **3,4,1,2**. Their four-score records are respectively (2,2,2,2), (1,1,1,1), (4,4,4,4), (3,3,3,3). Any one score distinguishes these four, though these are not all 32 possible secrets. Revisit the hidden games using the counts; then offer the general two-case proof from page 4 of this guide.

Keep the research conclusion precise: four fixed probes suffice to distinguish all 32 five-bit words; this packet does not prove a five-place optimum. The hypercube/landmark translation belongs in the discussion after decoding works. Binary distance is number of mismatches, hence five minus score.

# Games are recursive mathematics

**Question:** Which player has a strategy against every reply? Can a finite chart justify an infinite pattern? What information is needed when two games are combined?

## Prepare and launch

Bring 25 counters per group, large move cards **1 or 2**, **1, 3, 4**, and **1, 3, 5**, and pencils. Start with page 1 for each child (seven student sheets). Keep the remaining pages as continuation masters, including any earlier records children will compare. Print the extra only when selected. Pre-numbered charts reduce copying. Two distinct mats identify the extra's two piles. For the memory reserve, bring a paper strip and a movable four-cell frame (or two folded scraps).

Let children move and count counters before rules. Demonstrate 1-or-2 from three counters; let a child move and check legality. Remove an allowed positive number no larger than the pile. Taking the last counter wins. Older children then receive the clearly different 1,3,4 card. Do not announce the losing-number pattern at the launch.

K: count/remove 1 or 2 and alternate turns; no reading; examine both replies. Middle: counts and subtraction to 14 (physical allowed); adult can read; distinguish one winning move from all losing options. Upper: subtraction/remainders to 28, with groups of seven as a concrete alternative; two-direction argument and induction. Extra: small nonnegative labels, smallest-missing operation, and decreasing two-pile games; no binary arithmetic needed for the printed proof.

## A shared hour with seven children and two adults

0–10: explore materials freely. 10–15: brief launch and rules. 15–35: main investigation. 35–40: movement/reset; preserve work. 40–55: continue or choose an optional proof. 55–60: share and tidy. This flexible default is our proposal, not a source prescription.

One adult stays with K,K,1. The organizer alternates middle and upper, leaving a concrete next attempt and returning for an explanation. In games keep two opposing sides and rotate a checker. Proof continuations need adult attention; pages are not a completion quota.

**K-1 stop:** Demonstrate both replies that make three a trap. **Optional proof:** group counters in threes and explain the strategy for all multiples of three.

**Middle stop:** Defend W/L labels through seven. **Optional proof:** refute greedy play at eight and justify both winning moves at ten.

**Upper stop:** Conjecture losing remainders 0,2 and defend sample labels. **Optional proof:** show both remainder obligations and termination; reserve the paper-window argument for a later step.

# Endgames build the whole strategy

## K-1: groups of three

From 3, go second: answer 1 with 2 or 2 with 1. From 6, do the same, leaving 3; repeat. From 7 take 1 to leave 6; from 8 take 2; from 9 go second. Multiples of 3 are losing for the next player. Each pair of turns removes one group of three; the second player takes the last. Two demonstrated replies are a proof without a formula.

## Middle: backward reasoning

For moves 1,3,4, positions 0–20 are:

|      |    |    |    |    |    |    |    |
|------|----|----|----|----|----|----|----|
| Pile | 0  | 1  | 2  | 3  | 4  | 5  | 6  |
| W/L  | L  | W  | L  | W  | W  | W  | W  |
| Pile | 7  | 8  | 9  | 10 | 11 | 12 | 13 |
| W/L  | L  | W  | L  | W  | W  | W  | W  |
| Pile | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| W/L  | L  | W  | L  | W  | W  | W  | W  |

Zero is L for the player who would move next. If a position has a move to L, take it: the position is W. If all moves go to W, each gives the opponent a winning strategy: it is L. Every move decreases the pile, so this bottoms out. It is backward induction, not a tally of who happened to win.

From 5 take 3 to leave 2; from 6 take 4. From 7 all moves leave 6,4,3, which are W. Taking 3 from 8 leaves 5, and the opponent can take 3 to leave 2. Take 1 instead, leaving 7. From 10, taking 1 (leave 9) or 3 (leave 7) wins. Both are winning choices; taking 4 leaves W position 6.

## Hints that preserve investigation

Ask about one or two counters. For a disputed label, lay out all legal next piles. Check whose turn the label describes. If a child calls 7 winning because they won once, replay as the opponent with correct replies; return the discovery to the child.

**Returns:** Lowell Handout 2.3 and Handout 3.1–3.3 used 1-through-3 and 1-through-5. Those establish previous materials, not mastery. Handout 3.1 mistakenly calls 5 losing after observing it can move to losing 4; with those old rules, 5 is winning. Use the checked definitions here.

# An infinite proof from finite structure

## Upper: losing remainders 0 and 2 modulo 7

Proposed L piles are  $7k$  and  $7k + 2$ , including zero. Subtraction of 1,3,4 never connects these remainders. Every other remainder has a legal move into the set:

|               |   |   |   |   |   |
|---------------|---|---|---|---|---|
| Remainder     | 1 | 3 | 4 | 5 | 6 |
| Take          | 1 | 1 | 4 | 3 | 4 |
| New remainder | 0 | 2 | 0 | 2 | 2 |

Each move is legal even for the smallest positive representative. These are both proof obligations. Starting outside the set, move into it; every opposing reply exits; move back. The decreasing pile ends at zero, reached by your move. Starting inside gives the second player this strategy. The argument covers all piles, not just checked examples.

## Change to 1,3,5

Every move is odd and changes parity. No move connects even piles. Every positive odd pile can take 1 to reach even. Even piles are L and odd piles W: second wins from 6 or 100, first from 7. The proof structure persists while the invariant changes. This is different from taking any number 1 through 5.

## Why finite subtraction games eventually repeat

Copy labels onto a straight strip and frame four consecutive entries. To add the next label, inspect entries one, three, and four places back: the rightmost, second-from-left, and leftmost cells. Slide the frame one cell right after adding it. Match two windows and physically extend both to see the same next label. For 1,3,4, once four labels exist, the next depends only on those four. There are at most  $2^4 = 16$  W/L windows. Among 17 windows some pair agrees. Identical windows give identical next labels; induction forces all future labels to repeat. This proves *eventual* periodicity, not repetition from zero or period seven. The remainder proof supplies those stronger facts here.

For any finite move set with maximum  $m$ , use  $m$ -label windows, starting after enough entries exist to use the full recurrence. Grundy values (next page) are bounded by the number of allowed moves, so their windows also eventually repeat. Infinitely many allowed moves do not provide a fixed finite-memory argument.

## Adult attention and optional depth

The oldest child should defend a disputed label and a strategy to an adult. If the remainder investigation takes the whole session, save the finite-window argument and extra. Record which proof actually appeared; a completed chart alone is not understanding why a pattern continues.

# Beyond W and L: combining games

## Extra: smallest-missing labels

For 1,3,4, labels for piles 0–10 are **0,1,0,1,2,3,2,0,1,0,1**. Thus (1,1) loses; (1,4) wins by taking 1 from the second pile to leave (1,3). Both (1,3) and (4,6) lose; (5,6) wins by taking 1 from the first pile to leave (4,6). W/L status alone forgets information needed when games combine.

**The two-pile proof:** By the smallest-missing definition, no option from label  $a$  has label  $a$ , and every smaller nonnegative label occurs among its options. A move from equal labels changes exactly one, destroying equality. With unequal labels, change the larger to the smaller; the definition guarantees such a move. Both losing-position obligations hold. Total size decreases, and terminal (0,0) has equal labels. This is the two-component case of Sprague–Grundy theory. For three or more piles the rule is bitwise XOR.

## Undergraduate and research lineage

Ferguson’s UCLA Math 167 course includes take-away, Nim, graph games, and sums. His *Game Theory*, Part I, §1.3–1.4 develops winning/losing and subtraction games; §3.2 defines Sprague–Grundy values and §4.2 treats sums. Our proofs are concrete instances.

Shenxing Zhang, *On the linearity of the periods of subtraction games* (2024), §1, Lemma 1.1 and Conjectures 1.2–1.3, investigates changes in periods and preperiods when one move is added to a fixed set; Theorem 1.4 proves the predicted behavior in several families. Our seven-period example is elementary; predicting whole families is research. We describe the paper’s conjectures, not an exhaustive survey of their current status.

## Sources, reserves, and verification

JRMF *Countdown Teaching Guide*, PDF pp. 1–2: play-first questions; pp. 4–5: W/L reasoning; p. 8, challenge 4 explicitly uses 1,3,4. Our parity comparison, finite-window scaffold, and extra extend it. *Math Circle by the Bay*, printed pp. ix–x supports manipulatives and local extensions; the two-adult arrangement is ours.

Use F07-K-v3/M-v3/U-v3/X-v3, prepared, not taught. Save misère play, other move sets, and three-pile XOR for later. `plans/verify-week-07.py` checks 1,001 single-pile values and independently solves two-pile games with each pile at most 30; proofs above establish the unbounded claims.

# Stages and new concrete checks

## Revision v3: unpiloted

No Week 7 session evidence is supplied. This revision applies the Week 1 feedback by putting repeated play and legal-option records before W/L, grouped counters before remainders, actual window extensions before a finite-state proof, and two-pile games before number labels. These design choices await classroom evidence.

After handling the counters, gather everyone for 1-or-2 play from three. Let a child make one legal move, say the number taken, and record the number left. Check the final move. Then distribute first pages. Introduce older groups' 1,3,4 move card explicitly. A win in one game is an observation, not a guarantee against every reply. Demonstrate W/L only when a child has met that distinction in small endgames.

## K and middle

K page 1 records actual games. Page 2's replies are 2 after 1 and 1 after 2, from either 3 or 6; from 6 they leave 3. Page 3: choose first from 7 (take 1), first from 8 (take 2), second from 9. From 12, paired-turn replies leave **9,6,3,0**. The general groups-of-three argument remains an oral follow-up, not a required written conclusion.

Middle page 1's pile 2 can only leave 1; pile 4 can leave 3,1,0. Page 2 builds the labels through 7 with complete legal destinations. Stop there if all-versus-one reasoning still needs play. Page 3: from 8 the options are 7L,5W,4W; from 10 they are 9L,7L,6W. Therefore 8 has just the winning take-1 move; 10 has winning takes 1 and 3. For self-chosen strategy cards through 14, use the checked chart on page 2 of this guide.

## Upper: each representation has a task

Page 2 Problem 4: 14 reaches 13W,11W,10W; 16 reaches 15W,13W,12W; 15 reaches 14L,12W,11W; 17 reaches 16L,14L,13W. Page 3 builds these as two sevens plus leftovers 0,2,1,3. The remainder table has L at 0,2. From 0, subtraction of 1,3,4 reaches remainders 6,4,3; from 2 it reaches 1,6,5. W remainders have the winning moves on this guide's page 3. From 24 take 1 to leave 23, a losing pile; later replies maintain remainder 0 or 2. Discuss both losing-position obligations and legal boundary cases after the table, preserving the full proof.

Page 4's changed game has even losing piles. In the old game's windows, starts 0 and 7 read LWLW; next is W and the slide gives WLWW. Starts 1 and 8 read WLWW; next is W and the slide gives LWWW. Other valid matching windows differ by seven in the displayed chart. Only after these actual extensions introduce 16 possible windows and the eventual-periodicity argument. It does not by itself prove period seven or repetition from zero.

## Extra: use the labels after meeting the problem

Page 1 outcomes: (1,1) L, (1,3) L, (1,4) W, (4,4) L. From (1,4), the winning first moves take 1 or 3 from the pile of 4 to leave (1,3) or (1,1). Page 2's g-values are on this guide's page 4. From pile 4, next labels are 1,1,0 (missing 2); from 5 they are 2,0,1 (missing 3). Page 3 adds (3,4), won by taking 1 or 3 from the pile of 4 to equalize labels.

For (4,6), first moves from pile 4 of 1,3,4 can be answered from pile 6 by 3,3,4 respectively. First moves from pile 6 of 1,3,4 can be answered by: take 1 again from that pile; take 1 from pile 4; take 4 from pile 4. Each restores equal labels. Then offer the general mex/equality proof; do not infer it solely from these examples.

Start with seven first-page sheets and continuation masters. Keep earlier charts with later pages; stop after a defended small strategy, one general argument, or a tested label comparison. Record actual moves, misunderstood rules, adult rescue, voluntary continuations, and claims justified, separating observations from explanations. Flexible pace and action before tables are supported by *Math Circle by the Bay*, pp. ix–x, and Rozhkovskaya, Lessons 3,7,8; this sequence is our adaptation.

# From play to a useful record

F08 v3, September 2026: prepared and unpiloted. These changes apply the organizer's Week 1 feedback; there is no reported Week 8 classroom outcome. The goal is a strategy discovered through repeated concrete games. Pages are a menu for more than one possible meeting.

## Prepare a small starting set

Print page 1 for each child at their starting level: three K–1, three middle, one upper (7 sheets). Keep one master of each continuation and the 3-page extra; copy only what is needed. Middle page 3 reuses its earlier board. Bring about 60 counters, five plates or marked pile areas, pencils, and scraps labeled 1,2,4,8. Printed pile mats are usable counter areas; reset and reuse them.

K–1: adult reads; compare distances/count to 3. Middle: count to 7 and track moves, with adult scribing if needed. Upper: small subtraction and odd/even; bundles introduce binary notation. Extra: organize lists and compare differences; use a calculator only for the final supplied formula. Choose by prerequisites, not grade.

## Common launch before separate starts

0–10: let everyone move counters and build piles. 10–15: gather at one visible rook board. Show a one-square right move, a longer down move, and the same token reaching the star. Invite a child to make a legal move; have another mark its start and finish. Say: “One token, right or down, at least one square; reaching the star wins.” Show an illegal diagonal only if needed. Do not reveal equal distances yet. Place the token at (2,1) and build matching piles of 2 and 1. Slide right one square while removing one counter from the 2-pile. State that pile games remove any positive number from just one pile; taking the last counter wins. Then distribute starting tasks. This one correspondence shows both actions without revealing a strategy.

15–35: play the contrasting starts. 35–40: movement/reset. 40–55: repeat, compare, or introduce the next record only when it answers the children's question. 55–60: share one move with its reason and tidy. The timetable is an adaptable proposal, not a source-prescribed duration.

Parent stays mainly with K–1. Organizer alternates middle/upper about every five minutes, leaving a specific next game. Triplets rotate two opponents and a checker. The upper child needs actual games and discussion with an adult.

## Pacing, hints, and stopping

Ask for a legal move before a strategy. Then ask what is left to the star, or compare an equal and an unequal start. K can stop after showing the two center moves. Middle can stop with an argued board pattern or the board/pile correspondence. Upper can stop after the six replies to (1,2,3), or after balancing concrete bundle records. No general proof is a printed completion requirement; pages 2–3 retain the proofs for conversation. The extra can stop after its small map.

# Checked two-pile reasoning

## K-1 and middle board answers

Distances  $(a, b)$  mean squares remaining right and down. A rook move reduces exactly one positive coordinate. The star is  $(0, 0)$ : the previous player has already won. On the  $3 \times 3$  board, the center  $(1, 1)$  is losing. Both possible moves allow an immediate reply to the star. Top-left  $(2, 2)$  is also losing; copy the amount removed in the other direction.

On the  $4 \times 4$  board, the traps form the top-left to bottom-right diagonal:  $(3, 3), (2, 2), (1, 1), (0, 0)$ . Every unequal pair is winning: reduce the larger coordinate to the smaller. The middle start  $(3, 2)$  has the unique winning move to  $(2, 2)$ .

## A complete proof, with counters

If two piles have the same size, every move makes them unequal. If they differ, reducing the larger to the smaller is legal and restores equality. Thus a player who hands over equal piles can always restore equality after an opponent's move. The total number of counters strictly decreases, so play must finish. The player restoring equality takes the final counters; the opponent receives  $(0, 0)$  and cannot move. These two properties establish all traps, on rectangles as well as squares.

Any unequal two-pile start has exactly one winning first move (specifying the pile and amount). Matching piles have none. Children can use  $(5, 2)$  as their designed puzzle: remove 3 from the 5-pile.

## Why three piles change the problem

$(1, 1, 1)$  is winning: remove one pile and leave  $(0, 1, 1)$ . Even total is insufficient:  $(1, 1, 2)$  has total 4 but is winning by removing the whole 2-pile.  $(1, 2, 3)$  is losing. Here are all first moves and one reply; pile order is kept fixed.

| After move  | Reply leaves | After move  | Reply leaves |
|-------------|--------------|-------------|--------------|
| $(0, 2, 3)$ | $(0, 2, 2)$  | $(1, 2, 0)$ | $(1, 1, 0)$  |
| $(1, 0, 3)$ | $(1, 0, 1)$  | $(1, 2, 1)$ | $(1, 0, 1)$  |
| $(1, 1, 3)$ | $(1, 1, 0)$  | $(1, 2, 2)$ | $(0, 2, 2)$  |

Each reply leaves two equal piles and one empty pile. This gives a physical proof before binary notation. The middle group can stop here or join the upper bundle experiment.

# The binary invariant

## Upper answers

Binary records:  $1 = 001$ ,  $2 = 010$ ,  $3 = 011$ ,  $4 = 100$ ,  $5 = 101$ ,  $6 = 110$ ,  $7 = 111$ .  $(1, 2, 3)$  and  $(2, 4, 6)$  have even column sums.  $(1, 2, 4)$  does not; reduce 4 to 3. For the table:  $(2, 4, 6)$  is balanced;  $(3, 4, 5)$  can become  $(1, 4, 5)$ ;  $(1, 5, 7)$  can become  $(1, 5, 4)$ ;  $(3, 5, 6)$  is balanced. Other correct balancing moves should be accepted.

For  $(7, 10, 12)$ , the Nim sum is 1; reduce 7 to 6. For chosen first piles  $a, b$ , the unique third pile is  $a \oplus b$ : each binary digit is forced by whether the first two digits agree.

## Why the test is sufficient, for any number of piles

A move changes exactly one integer. At least one of its binary digits changes, so at least one previously even column becomes odd. No legal move can connect two balanced positions.

If the Nim sum  $s$  is nonzero, choose its highest 1-bit, say the  $2^k$  column. Some pile  $x$  has a 1 there. Replace it by  $y = x \oplus s$ . Higher bits stay fixed, this bit becomes 0, and lower bits may change. The sum of all smaller bundles is  $2^k - 1$ , so  $y < x$ : the move is legal. Every odd column is toggled once, every even column is unchanged, and the result is balanced. The decreasing total ensures termination. These two facts prove exactly the same win/lose structure as in the two-pile proof.

Do not physically permit removing an arbitrary bundle while leaving a larger number of counters: the game is still ordinary removal from one pile. The bundle display is a way to calculate a legal smaller result.

## Extra: Wythoff, checked results and proof boundary

The old equal-pile trap fails because both piles can be emptied in one diagonal move. For  $0 \leq a \leq b \leq 7$ , the traps are  $(0, 0)$ ,  $(1, 2)$ ,  $(3, 5)$ ,  $(4, 7)$ . Greedy pairs through  $n = 5$  are  $(0, 0)$ ,  $(1, 2)$ ,  $(3, 5)$ ,  $(4, 7)$ ,  $(6, 10)$ ,  $(8, 13)$ .

Distinct pairs have disjoint coordinates and distinct absolute differences, so neither a single-pile move nor an equal-take move connects them, including reversed pile order. This proves only the no-move-between-traps condition. To prove completeness one must also show every outside position reaches a listed pair. Sample moves:  $(2, 4) \rightarrow (2, 1)$ ,  $(5, 8) \rightarrow (4, 7)$ ,  $(8, 12) \rightarrow (6, 10)$ . The golden-ratio formula is a cited theorem, not a conclusion from five data points.

# Where the mathematics leads

## Undergraduate direction and genuine research

Two piles give a strategy-preserving change of representation from a grid to an impartial game. Binary column parity is addition in a vector space over the two-element field. The proof has a useful algorithmic form: recognize the losing set, prove no move stays inside it, and construct a move into it from everywhere else. Formal algebra vocabulary can wait.

Charles Bouton's historical paper, *Nim, a Game with a Complete Mathematical Theory*, *Annals of Mathematics* 3 (1901–02), pp. 35–39, is the original research source for binary analysis. Our normal-play proof is written out on page 3, so the circle does not depend on reading the historical notation. Caltech-hosted paper.

Fraenkel and Peled, *Harnessing the unwieldy MEX function*, *Games of No Chance 4*, MSRI Publications 63 (2015), pp. 77–94: §1, pp. 77–78 states the Wythoff rule, greedy pairs, and golden-ratio formula; §5 develops efficient computation for generalized complementary sequences. Author research chapter.

The extra packet meets an actual research theme: a simple rule change can replace bitwise arithmetic with complementary sequences and a question of efficient computation. It does not reproduce the paper's generalized algorithm or claim an open problem has been solved by the classroom experiment.

## Local sources and teaching choices

JRMF, *Rook's Move Activity Guide*, PDF pp. 3–6: legal moves, exploration before advice, losing positions, and the Wythoff connection. We replace the old upper king variant with three-pile Nim; the king version remains an untaught reserve unless the use log says otherwise.

*Math Circle by the Bay*, preface printed pp. viii–x (PDF pp. 9–11), supports common themes with adjustable depth, manipulatives, explanations, and extra challenges. Rozhkovskaya, Lesson 7 “At the lesson,” EPUB part0017.xhtml, highlights checking that children understand legal moves. The seven-child schedule and bundle prompts are our adaptations.

Lowell Handout 2, problem 2.3, and Handout 3, problems 3.1–3.3, include subtraction-game history; Week 7 also studies constrained subtraction. Here an entire pile may be removed, and the important new step is multiple piles plus binary balance. Reuse is deliberate, not evidence that every child has seen these exact starts.

## Record and verify

IDs F08-K/M/U/X-v3. Record the actual starts, rule, level, strategies explained, and whether Wythoff was attempted. Keep unused stages for later years. `verify.py` checks all three-pile states through size 12 and Wythoff states through 30 by independent backward recursion. Those finite checks support the examples; the proofs establish the unlimited claims.

# Use each record for a reason

## Core sequence and specific hint ladders

**K pages 1–2:** compare an immediate win with the center, then equal top-left distances with an unequal start. Demonstrate drawing a move from square center to square center. When stuck, enumerate the center's two legal moves with the token. Page 3 is a separate translation: build two equal piles, copy a removal, then change to unequal piles. Stop before this transition if rook play remains engaging.

**Middle pages 1–2:** play eight purposeful starts before the T/W map. Demonstrate the (right-distance, down-distance) record at (2,1) with the actual token. The first table records game outcomes, not proved winning types. Before mapping, classify (1,0) and (1,1) together: a W has one move to T, a T has every move to W. Ask children to challenge one proposed letter; a lost game alone proves neither T nor W. Page 3 physically matches two slides to removals before asking for equal-pile replies. Page 4 adds three piles only after equal-pile play is understood. The six preprinted outcomes prevent copying from consuming the investigation.

**Upper pages 1–2:** compare adjacent starts and test the even-total guess. Two equal piles plus an empty pile is useful, but (2,4,6) has no immediate move to that form: failing to find one is a reason to seek another representation. The six replies to (1,2,3) can be a complete stopping point before binary. Page 3 groups real counters before recording 0/1 columns. Model 5 as 4+1, then let the child build. If notation obscures the game, leave bundle cards visible beside ordinary piles. Page 4 tests the same record on four cases and actual opponent moves. Page 5 introduces the highest-odd-column recipe only after the repair problem is familiar.

**Proof reserve:** ask why a move from balance must alter a column; then why the highest changed bundle outweighs all lower bundles together; finally use decreasing total for termination. Preserve both directions of the losing-set argument. Naming XOR is optional and follows the child's record, not the launch.

## Extra and observation record

Play equal and unequal starts before the classification grid. Work only far enough to reveal a second nontrivial trap; finish the grid later if useful. Demonstrate sorting after a move: (2,4) can become (2,1), recorded at (1,2). The greedy sequence is a compressed record of already encountered traps. Distinct coordinates/differences prove no internal moves; completeness is a separate requirement. Supply the golden-ratio result as a theorem only if reached.

After teaching, record exact starts/pages, who needed a rule repeated, useful or burdensome records, child conjectures, and requests to continue. Separate observations from hypotheses and adult-supplied theorems. Mark unused pages unpiloted; do not infer uptake from printing.

# Additional v3 solution checks

## Board and pile instances

K1 immediate-above-star is (0,1), winning immediately; the center is (1,1), losing. From top-left (2,2), right 1 is answered down 1; right 2 by down 2, and vice versa. The middle-left start (2,1) can move right 1 to (1,1). Equal pile starts (2,2),(3,3) use matching replies; (3,1) removes 2 from the larger pile.

Middle trial starts (1,0),(2,1),(3,1),(3,2),(2,3) are W; (1,1),(2,2),(3,3) are T. Unique moves to equal distances respectively leave (0,0),(1,1),(1,1),(2,2),(2,2). At (3,2), right 1 corresponds to piles (2,2); down 2 to (3,0). In pile Problem 6, (3,2) reduces to (2,2), (1,4) to (1,1); equal starts require an opponent's move first. The six replies in middle Problem 8 and upper Problem 4 are listed on page 2.

## Upper bundle work

Problem 3: (1,1,1) removes any single 1; (1,1,2) removes the 2. Neither (1,2,3) nor (2,4,6) reaches two equal nonempty piles and an empty pile in one move, since no pair is initially equal. Their totals 6 and 12 are even, but the total-4 winning example (1,1,2) refutes even-total sufficiency.

Problem 5 records in 4/2/1 order are 1=001, 2=010, 3=011, 4=100, 5=101, 6=110, 7=111. Problem 6 column counts for (1,2,3) are (0,2,2); for (1,2,4), (1,1,1). Reduce 4 to 3. Problem 8 checked moves are on page 3; multiple balancing moves can occur.

Problem 10: (4,6,7) has odd 4 and 1 columns. Legal repairs are 4 to 1, 6 to 3, or 7 to 2. In (7,10,12), only the 1 column is odd; reduce 7 to 6. Problem 11 third piles are 7 for (2,5) and 5 for (3,6); each bit is forced. Accept a child-designed zero-XOR start with a demonstrated legal reply.

## Extra instances

Problem 1: (2,2) and (2,3) are winning, moving respectively to (0,0) and (1,2); (1,2),(3,5) are traps. Problem 2's four outcomes are (0,2),(1,1),(1,0),(0,1); empty the remaining pile or take 1 from both. Problem 3 grid has exactly (0,0),(1,2),(3,5),(4,7) as T. Problem 4: (2,4) to (2,1); (4,6) to (3,5); (5,7) to (3,5), all legal. Greedy/formula rows and final large moves are on page 3. For each tested pair, single-pile removal would need a shared coordinate; equal removal would need the same difference. Neither occurs between distinct listed traps.

# Bounces before mirrored rooms

F09 v3, September 2026: prepared and unpiloted. Applying Week 1 feedback is a design inference, not evidence these billiards pages have been taught. Tracing comes before unfolding; unfolding gets its own concrete work before arithmetic formulas.

## Preparation and access

Print page 1 at each child's starting level (3 K–1, 3 middle, 1 upper: 7 sheets), with continuation masters and the 3-page extra available. Copy K page 3 separately when folding; cut between its two panels first. Keep earlier traced boards for later comparisons. Bring counters, pencils, rulers, scrap grid paper, and scissors for an adult. About 10 minutes' preparation; no actual ball.

K: adult reading/tracing allowed; child chooses turns. Middle: count lengths and multiples through 12, then count whole rooms. Upper: multiples, division and parity; gcd/lcm and the coprime lemma are optional adult-supported continuations. Extra: fractions and coordinates; supply irrationality if necessary. Oral reasoning and adult scribing are valid.

## Common launch before separate starts

0–10: let children handle counters and trace grid diagonals freely. 10–15: gather at a visible  $2 \times 4$  table. Move diagonally from the black dot to the right wall. Ask a child to show the next move that stays inside and still heads upward. Dot that bounce and circle the ending corner. Show that interior line crossings do nothing. Everyone now knows the physical action and what to record; distribute different starts only afterward. Save mirror rooms for a later question about predicting a path.

15–35: trace and compare. 35–40: movement/reset. 40–55: revisit a puzzling path, fold one copy, or continue the room-count investigation. 55–60: share one checked prediction and tidy. These times are adjustable. Parent mainly supports K; organizer alternates middle/upper visits and leaves a specific next rectangle. Rotate mover, tracer, checker, so each child acts.

## Hints and useful stopping points

First check the bounce rule. Then enlarge or trace a difficult section while the child directs. Introduce mirrored rooms by physically folding a line across one wall, not by announcing the lcm. K may stop with contrasting corners. Middle may stop with matched bounce/crossing numbers, then later predict corners. Upper may stop with room-count parity and a verified bounce rule. General coprime division, no-bottom-left and complete inverse-design proofs remain adult conversations on pages 2–3. A returner can supply one familiar trace and proceed to the representation they have not used.

# Checked paths and corner rule

## K-1 and middle answers

For width  $w$ , height  $h$ :  $2 \times 2$  ends top-right with no bounce.  $2 \times 4$  follows  $(0, 0)$ ,  $(2, 2)$ ,  $(0, 4)$ , ending top-left;  $4 \times 2$  follows  $(0, 0)$ ,  $(2, 2)$ ,  $(4, 0)$ , ending bottom-right. Folding the two width-2, height-4 rooms turns the straight diagonal through  $(2, 2)$  into the first of these paths.

For  $2 \times 3$ , the folded path is  $(0, 0)$ ,  $(2, 2)$ ,  $(1, 3)$ ,  $(0, 2)$ ,  $(2, 0)$ . In copied rooms, the first simultaneous wall meeting is  $(6, 6)$ . The four middle rows have  $(L, \text{widths}, \text{heights}, \text{corner})$ :

| Table        | $L$ | Widths | Heights | Corner       |
|--------------|-----|--------|---------|--------------|
| $2 \times 3$ | 6   | 3      | 2       | bottom-right |
| $3 \times 6$ | 6   | 2      | 1       | top-left     |
| $3 \times 9$ | 9   | 3      | 1       | top-right    |
| $4 \times 6$ | 12  | 3      | 2       | bottom-right |

A middle design reaching top-right after a bounce can be  $2 \times 6$  (two bounces); top-left can be  $2 \times 4$ . Accept any verified design.

## Upper theorem with all hypotheses

Take positive integer  $w, h$  and a 45-degree launch. Reflections unfold to  $y = x$ . A corner occurs when the common coordinate is both a multiple of  $w$  and of  $h$ . Its first positive value is  $L = \text{lcm}(w, h)$ . First use room counts  $m = L/w, n = L/h$ . If they shared  $d > 1$ , distance  $L/d$  would already cross  $m/d$  widths and  $n/d$  heights: an earlier corner. Thus they are coprime; both cannot be even, ruling out bottom-left without a gcd formula.

**Optional arithmetic bridge.** Write  $w = ga, h = gb$  with coprime  $a, b$ . Any common multiple is  $gat$  with  $b \mid at$ . To justify  $b \mid t$ , begin with  $a = 2, b = 3$ : both  $3t$  and  $2t$  divide into threes, so their difference  $t$  does too. In general, run repeated subtraction on strips of lengths  $a, b$  and, alongside them,  $at, bt$ . Subtract the shorter from the longer until equal. Common divisors are unchanged, so the coprime base strips end at 1 and the scaled strips at  $t$ . Both scaled lengths remain multiples of  $b$  throughout; hence  $b \mid t$ . The least positive  $t$  is therefore  $b$ , and  $L = gab, m = b, n = a$ . The shared plan scaffold supplies physical bundle examples. If this step is supplied, record it as a given theorem.

An odd number of whole widths folds to the right, even to the left. An odd number of whole heights folds to the top, even to the bottom. Thus odd  $a$ , odd  $b$  gives top-right; even  $a$ , odd  $b$  gives bottom-right; odd  $a$ , even  $b$  gives top-left. Both cannot be even, so first arrival at bottom-left is impossible. Scaling both dimensions preserves the reduced pair and corner.

Upper rows:  $4 \times 6$ :  $L = 12, m = 3, n = 2$ , bottom-right;  $6 \times 9$ : 18, 3, 2, bottom-right;  $6 \times 10$ : 30, 5, 3, top-right;  $8 \times 14$ : 56, 7, 4, bottom-right. Raw parity of the original lengths does not determine the endpoint.

# Bounces, inverse design, and slopes

## Count crossings, then design

Before the final corner the line crosses  $m - 1$  vertical room walls and  $n - 1$  horizontal ones. With the optional lemma these are  $b - 1$  and  $a - 1$ . No crossing belongs to both lists: that would be an earlier corner. So the number of bounces is  $a + b - 2$  (not counting launch or terminal corner). Path length, if wanted, is  $\sqrt{2}L$ ;  $L$  itself is horizontal distance, not path length.

For top-right after exactly four bounces,  $a, b$  are coprime odd positives with  $a + b = 6$ . Candidates  $(1, 5), (3, 3), (5, 1)$  reduce to just  $(1, 5), (5, 1)$  because  $(3, 3)$  is not coprime. Any positive integer scaling works, such as  $2 \times 10$ . Top-right after three bounces is impossible: two odd numbers have even sum, so  $a + b - 2$  is even. The latter is a parity obstruction, not a failed search.

## Extra packet: rational and irrational directions

The extra uses a *unit square* and physical slope  $s = p/q$  in lowest terms. A positive lattice point  $(u, v)$  on  $y = (p/q)x$  satisfies  $qv = pu$ , so its first occurrence is  $(q, p)$ . By the coprime lemma on page 2,  $q$  divides  $u$ ; the first positive choice is  $u = q$ , giving  $v = p$ . The endpoint's horizontal side is right iff  $q$  is odd; its vertical side is top iff  $p$  is odd. There are  $p + q - 2$  bounces. Slope  $2/3$  ends bottom-right with 3 bounces;  $3/2$  ends top-left with 3.

For slope  $\sqrt{2}$ , an integer corner  $(u, v)$  with  $u > 0$  would give  $\sqrt{2} = v/u$ , impossible. If requested, prove irrationality: reduce a supposed fraction  $v/u$  first. From  $v^2 = 2u^2$ ,  $v$  is even; substituting  $v = 2k$  forces  $u$  even, contradicting the reduction.

The other launch  $(0, 1/2)$  with slope 1 cycles through  $(1/2, 1), (1, 1/2), (1/2, 0), (0, 1/2)$ . It avoids corners but follows just a diamond. No-corner does not imply dense. Density of the  $\sqrt{2}$ -slope trajectory launched from  $(0, 0)$  is a separate theorem. Under our stopping rule, arbitrary starts need care: from  $(0, 2 - \sqrt{2})$  the same slope reaches unfolded  $(1, 2)$ , folds to bottom-right, and stops.

## Keep the conventions straight

A general rectangle with physical slope  $s$  becomes a square after rescaling axes, with normalized slope  $sw/h$ . Rationality of that normalized slope controls the square model. The extra fixes  $w = h = 1$  to avoid silently mixing those slopes. "Rational polygon" in research means its angles are rational multiples of  $\pi$ ; it does not mean a rational launch slope. Our corner-start paths stop when they hit a corner, so do not call those stopped paths periodic billiard orbits.

# A doorway into dynamics

## Mathematical direction and primary sources

Unfolding replaces a piecewise reflected path by a straight line on a lattice. The corner problem is divisibility, coprime integers, and parity; the inverse problem asks for a complete classification of parameters. These are substantive undergraduate number-theoretic and geometric ideas in a tangible representation.

Masur and Tabachnikov, *Rational billiards and flat structures*, published in *Handbook of Dynamical Systems* 1A (2002), pp. 1015–1089. In the linked author manuscript, §1.3 gives unfolding, §1.4 develops the unit-square/torus model, and §1.5 extends the construction to flat surfaces. Later §4 concerns periodic orbits. Author manuscript.

The classroom genuinely performs the first construction used in the research: reflect copies, then study straight motion. It does not establish the deeper recurrence or ergodicity theorems. The extra explicitly separates avoiding corners from density; the latter is supplied specifically for the origin-start  $\sqrt{2}$  path under our stopping convention.

## Downloaded sources and documented prior use

JRMF *Billiards Geometry*, PDF p. 2, asks for corner, bounce-count, and path-length rules. Lowell Handout 10, problems 10.2–10.7, includes  $7 \times 5$ ,  $4 \times 3$ ,  $8 \times 10$ , scaling, consecutive sides,  $1 \times n$ , and  $2 \times \text{odd}$  families. Handout 11, problems 11.4–11.5, includes height-five rectangles and plus-shaped tables. Do not describe small rectangle families or scaling as wholly new for returning children.

Our changed emphasis is a proof through physical unfolding, an exact first-corner criterion, a bounce formula, and inverse existence/impossibility design. This deliberately promotes an old reserve into the core. Plus-shaped tables remain a later branch; do not claim the rectangle formula covers them.

*Math Circle by the Bay*, preface printed pp. viii–x, supports manipulatives and revisiting ideas at different depths. Rozhkovskaya, Lesson 3 “At the lesson,” EPUB part0013.xhtml, supports attempts and child explanations before formal displays; Lesson 8’s account warns about copying pace. Preprinted grids and optional adult tracing are our response, not a procedure claimed to be in those sources.

## Keep a useful history

IDs F09-K/M/U/X-v3. Record actual dimensions, launch slopes, whether unfolding was used, which formulas were explained, and which claims stayed conjectural. A prepared extension is not a taught activity. Future variants can change slopes, starting points, or polygons instead of enlarging the same arithmetic table.

`verify.py` independently traces all 900 rectangles with sides 1–30, compares the corner and bounce formulas, and checks the inverse-design list. The general proofs are on pages 2–3; finite checks alone do not establish them.

# Representation gates and records

## Move, mark, compare; then unfold

**K pages 1–2:** the square has no bounce, tall and wide tables turn at different wall types, and the height-6 table adds a second bounce. The child chooses directions even when an adult draws. Page 3 offers two separate folds, one across a vertical wall and one across a horizontal wall. Cut between the two diagrams; fold the right half left for Problem 5 and the top half down for Problem 6. A spare sheet or tracing paper makes the reflected segment visible. Folding is a follow-up to the paths, not an opening explanation.

**Middle:** page 1 gives three contrasting traces. Page 2 numbers bounces and unfolded crossings so the new picture has a concrete matching job. Demonstrate the first crossing (2,2); let the child match the rest. Page 3 gives separate 3-by-6 and 3-by-9 diagrams before translating parity into an endpoint table. Have the child actually move back and forth along a room edge as counts rise. Page 4 introduces lists of multiples to predict a larger table without drawing all copied rooms, then supplies design grids with stated bounds. No freehand rectangle is required to be metrically exact if the counted grid is correct.

**Upper:** page 1 contrasts a square and swapped side lengths. Page 2 gives original and copied 4-by-6 rooms side by side; the child matches actual bounces with crossings. Page 3 uses a scaled copy and two even-sided tables with different endpoints, preventing raw parity from being mistaken for the rule. The false distance 24 forces attention to the first corner. Page 4 counts separate wall lists before a formula. Page 5 compares three candidate designs, including the reducible 3-by-3 case. The two genuine four-bounce designs therefore contrast with an early-stop failure.

## Preserve the proof as a conversation

The least common distance argument, “earlier corner if both room counts share a divisor,” and wall-count proof can be shown directly on records. Only then offer reduced sides and the coprime-divisibility scaffold. If the child uses the formula but cannot justify its lemma, record it as supplied. The complete inverse-design proof and irrationality proof remain on pages 2–3. No child needs all these arguments in one hour.

The extra gives five rational trials, including  $2/4$  vs  $1/2$ , before the general  $p/q$  claim. The midpoint diamond gives a concrete counterexample to “no corner implies dense.” Keep the origin hypothesis on the supplied irrational-slope density theorem. Arbitrary offsets and general rectangles need the qualifications on page 3.

## After the meeting

Record actual tables/slopes, bounce-rule misunderstandings, whether folding clarified the path, which records were useful, and questions children wanted to pursue. Distinguish an observed difficulty from a proposed explanation; record conjecture, explained proof, and supplied theorem separately. These v3 changes remain unpiloted until actual use is reported.

# Additional v3 solution checks

## K and middle

K Problem 3: 4-by-2 ends bottom-right after 1 bounce; 2-by-6 follows (0,0),(2,2),(0,4),(2,6), ending top-right after 2. Problem 5 folds the straight 4-by-4 diagonal into the 2-by-4 path; Problem 6 folds it into the 4-by-2 path.

Middle page 1 corners/bounces: 2-by-2 top-right/0; 2-by-4 top-left/1; 4-by-2 bottom-right/1. Problem 4 unfolds 2-by-3 crossings at distances 2,3,4, ending at 6. The corresponding folded bounces are (2,2),(1,3),(0,2). Whole-room counts are 3 widths and 2 heights. Problem 6: 3-by-6 reaches (6,6), counts (2,1), top-left, 1 bounce; 3-by-9 reaches (9,9), counts (3,1), top-right, 2 bounces. Endpoint rows alternate right/left and top/bottom.

Problem 8: 4-by-6 has  $L=12$ , counts (3,2), bottom-right; 2-by-6 has  $L=6$ , counts (3,1), top-right; 3-by-3 has  $L=3$ , counts (1,1), top-right. Design examples within the 6-by-6 grids: 2-by-6 top-right after 2 bounces, 2-by-4 top-left after 1. A square fails the “after at least one bounce” condition.

## Upper

Problem 1: 2-by-3 ends bottom-right/3 bounces; 3-by-2 top-left/3; 3-by-3 top-right/0. Problem 4 for 4-by-6 has vertical distances 4,8,12 and horizontal 6,12: first common 12. The 6-by-9 scaling factor is  $3/2$ ;  $L$  scales 12 to 18 while counts and corner stay the same. 6-by-10 ends top-right; raw even side lengths do not determine corner. Problem 8's distance 24 predicts bottom-left, but the path stops at 12 first.

Problem 9: 2-by-3 vertical list 2,4, horizontal 3, total 3; square both empty, total 0; 4-by-6 lists 4,8 and 6, total 3; 6-by-10 lists 6,12,18,24 and 10,20, total 6. The count is  $(m-1)+(n-1)$ . Doubling dimensions doubles all distances, preserving counts. Problem 12: 1-by-5 and 5-by-1 have four bounces and end top-right; 3-by-3 stops immediately. Problem 13 may use 2-by-10. For Problem 14, three bounces would give  $m+n=5$ ; top-right requires both  $m,n$  odd, impossible. Page 3 gives the full reduced-pair classification.

## Extra

Slopes  $1/2$ ,  $2/3$ ,  $3/2$  first reach (2,1),(3,2),(2,3), with corners top-left, bottom-right, top-left and bounce counts 1,3,3. Slope  $2/4$  equals  $1/2$  and stops at (2,1), not (4,2). Slope  $3/4$  first reaches (4,3), top-left, 5 bounces. Problem 4 examples:  $1/5$  or  $5$  for top-right/4;  $2/3$  for bottom-right/3. For the midpoint diamond,  $(1/2,1/2)$  and a small neighborhood of it are never visited. Absence of corner hits alone does not establish density; the separate stated theorem does.

# Walk before counting degrees

F10 v3, September 2026: prepared and unpiloted. Week 1 feedback motivates more contrasting walks and records before parity, splicing or optimization. No Week 10 classroom outcome has been supplied. Maps are connected and undirected; parallel bridges are allowed, self-loops are not used.

## Preparation and access

Print page 1 for each child at the appropriate starting level (3 K–1, 3 middle, 1 upper: 7 sheets). Keep continuation masters and the 3-page extra; copy on demand. Retain earlier maps for comparisons. Bring seven moving counters, about 35 used-edge markers, pencils and scraps for route strips. Optional strings and paper islands enlarge a map; no calibrated printing is needed.

K: adult reads; distinguish an island from a bridge and track used bridges. Middle: counts to 4 and odd/even; adults may write route lists. Upper: follow route lists, pair arrivals/departures and compare constructions with lower bounds. Extra: add small lengths, compare three choices; shortest connections are a new idea introduced through actual walks.

## Common launch before separate starts

0–10: children freely build little maps or move along printed ones. 10–15: gather at a triangle. Walk a counter A–B–C–A, putting a marker on each used bridge. Say: “Visit every bridge once; revisiting an island is allowed.” Show how the route list records each visited island. Invite a child to try another start and mark a used bridge. Add a branch and start an attempt, then distribute starting pages. Save odd/even language for after contrasting attempts.

15–35: try walks and compare starts. 35–40: movement/reset. 40–55: repair a map, splice a second loop, or improve a delivery route at the children’s current stage. 55–60: share a route and a reason; tidy. Parent mainly stays with K; organizer alternates middle/upper about every five minutes. Triplets rotate walker, marker, and checker. These times are flexible proposals.

## Hints, gates and stopping

First ask which bridge is unused and where the counter is. Next compare a successful and a stuck attempt. Only then pair arrivals/departures. K may stop with two-leaf versus three-leaf attempts and one repair. Middle may stop with forced house endpoints or a repaired map. Upper may stop with two concrete splices; delivery becomes a new stage with a changed rule (repetition allowed). The full Euler construction and postman lower bounds are discretionary adult conversations on pages 2–3. Children need not finish every page to encounter substantial mathematics.

# One-walk maps: checks and proof

## K-1 answers

A triangle has a closed route, e.g.  $A-B-C-A$ . Add a branch  $A-D$ : a trail is  $D-A-B-C-A$ , with endpoints  $D$  and  $A$ . At a degree-1 leaf, a walk cannot arrive and then leave by a different unused bridge. Thus the three-leaf star cannot have a trail covering every edge once: it would need three endpoints, while a walk has at most two.

Adding a bridge  $B-C$  between two leaves repairs the star:  $D-A-B-C-A$  is a complete trail. It begins/ends at the remaining leaf  $D$  and center  $A$ . Accept another pair of leaves with relabeled route.

## Middle and upper map answers

House degrees:  $A=2, B=2, C=3, D=3, E=2$ . One trail is  $D-A-B-C-D-E-C$ . Every complete trail must start at one of  $C, D$  and end at the other. No closed Euler trail exists.

For the triangle with center  $D$ , all four vertices  $A, B, C, D$  have degree 3. It has no every-edge-once trail. Add a second  $AB$  bridge; now  $A, B$  are even and  $C, D$  odd. A valid trail is  $C-A-B-D-A-B-C-D$ , using the two distinct  $AB$  bridges separately. Middle children may draw the added bridge as a curve; a crossing without a dot is not a new junction.

## Euler criterion with a construction

In any completed trail, arrivals and departures at an internal vertex pair. A closed Euler trail therefore has only even degrees; an open Euler trail has exactly two odd vertices, its endpoints. The necessity argument includes repeated visits to a vertex: each visit contributes another pair.

First physically splice the two triangles  $ABC$  and  $ADE$  sharing  $A$ . Mark  $AB, BC, CA$  used after  $A-B-C-A$ ;  $AD, DE, EA$  remain. Insert the strip  $A-D-E-A$  at a visit to  $A$  to obtain, for example,  $A-D-E-A-B-C-A$ . Trace it and mark all six roads. This one construction motivates the next general argument.

For sufficiency, assume a finite connected graph with at least one edge and all degrees even. Follow unused edges until stuck. At a vertex other than the start, arriving uses one edge of an even available count and leaves at least one to depart; so only the start can be where the walk stops. If unused edges remain, connectedness ensures that some vertex of the current closed walk touches unused edges (take a path toward an unused edge). Start a new closed walk there in the remaining even-degree graph and splice it into the first. Repeat. Finitely many edges guarantee completion.

For exactly two odd vertices, add one temporary edge between them, construct the closed Euler trail, then remove that temporary edge to obtain an open Euler trail. Parallel edges are allowed. This proves existence; it does not promise that every arbitrary greedy walk uses all edges without splicing.

# Delivery routes and optimality

## Upper: six unit-length roads

The triangle-and-center network is  $K_4$ , with six unit roads. Every closed covering walk must repeat at least two road crossings: all four original vertices are odd, and each added traversal changes parity at its two endpoints only. A closed walk has even traversal-degree everywhere. Thus six original plus at least two repeats gives a lower bound of 8.

A matching route is A–B–C–D–A–C–D–B–A. It repeats AB and CD; all other roads occur once. The three possible odd-vertex pairings are AB+CD, AC+BD, AD+BC. Each adds two roads and creates an even connected multigraph, so each admits a closed Euler trail of length 8.

If finishing anywhere is allowed, at most two odd vertices may remain. At least one repeated crossing is necessary, giving a lower bound of 7. A–B–C–A–D–C–D–B attains 7, repeating CD.

For the design brief, a star with five leaves has six odd vertices: five degree-1 leaves and the degree-5 center. But its leaves are not adjacent, so it cannot meet the three-extra-step target. A workable design is two triangles joined by three corresponding cross edges (a triangular prism graph). All six degrees are 3. Duplicate those three cross edges; now all degrees are 4, so a closed tour exists with exactly three extra crossings, meeting the parity lower bound. Label the triangles ABC and DEF, with cross edges AD, BE, CF. One 12-step closed route is A–B–C–A–D–E–B–E–F–C–F–D–A. Children may find other designs.

## Extra: six roads of different lengths

Original total is  $2 + 3 + 7 + 1 + 6 + 1 = 20$ . Distances between odd vertices are  $d(A, B) = 2$ ,  $d(A, C) = 2$  via D,  $d(A, D) = 1$ ,  $d(B, C) = 3$ ,  $d(B, D) = 3$  via A,  $d(C, D) = 1$ .

Pair costs: AB+CD=3; AC+BD=5; AD+BC=4. Choosing the cheapest available AD first is not guaranteed optimal. Duplicate AB and CD to obtain minimum total 23. The same A–B–C–D–A–C–D–B–A route has length 23. Long roads AC and BD must still be serviced once; shortest connections are used only for the *extra* travel.

For a general lower bound, subtract one required copy of every road from a closed covering walk. The remaining multigraph has odd degree exactly at the original odd vertices. Its edges can be decomposed into paths pairing those odd vertices and closed loops: follow unused edges from an odd vertex to another odd one, remove that path, and continue, then remove cycles. Each path costs at least the shortest distance of its endpoint pair, and loops cannot improve a nonnegative-cost total. Thus every solution pays at least the best pairing cost. Duplicating shortest paths for that pairing attains the bound by Euler's construction.

# The mathematics behind the workshop

## Undergraduate graph theory

The optional proof conversation completes the Euler trail criterion, including its constructive direction; the physical splice is a satisfying earlier step. It distinguishes feasibility from optimization and separates an example route (an upper bound) from an obstruction (a lower bound). The upper activity turns parity into an algorithm for a small route-inspection problem.

Cornell CS 2800, Spring 2017, *Lecture 36: Paths*, gives the even-degree theorem and a cycle-splicing proof. Official course notes. Our maps, graded questions, and counter-based launch are original adaptations.

## A real research connection

Edmonds and Johnson, *Matching, Euler tours and the Chinese postman*, *Mathematical Programming* 5 (1973), pp. 88–124, studies the weighted route-inspection problem using matching theory. The introduction describes the problem and its structure; §4 concerns a matching algorithm and §5 Euler tours. IBM publication record; Paper hosted by University of Michigan.

The extra packet performs an exact four-odd-vertex instance of the same reduction. It does not teach the general blossom algorithm, polyhedral proof, or directed/mixed variants. For  $2k$  odd vertices, exhaustive pairings number  $(2k-1)(2k-3)\cdots 1$ ; efficient weighted matching makes large instances practical. The research is a historical solved result, not a claimed present-day open problem.

## Local sources and returners

Rozhkovskaya, *Math Circles for Elementary School Students*, M.4.1 “Make your own problem,” EPUB part0031.xhtml, inspires making under constraints. Its animal/figure/number prompt is different; our connected-map/parity specifications are our design. Lesson 3 “At the lesson” supports child attempts and explanations before an organizing display.

*Math Circle by the Bay*, preface printed pp. viii–x, supports adjustable depth and extra challenges. Our timetable and two-adult allocation are proposals for this roster.

Lowell Handout 10’s password-window puzzle is related to directed Euler/de Bruijn constructions; do not call graph traversal wholly new. Here the representation is an undirected bridge map, and the claims concern odd-degree obstructions and minimum repeated travel. Week 1’s flip graph concerned states and distance; here edges are roads that must all be serviced. Generic old workshop starters remain optional future reserves.

## Use history and checks

IDs F10-K/M/U/X-v3. Record maps, repairs, exact routes, whether optimality was justified, and child-designed problems with their solutions. `verify.py` independently searches states (current vertex, set of roads already visited) and confirms closed/open/weighted optima 8, 7, 23. It also checks printed routes. The mathematical certificates above explain why the results hold.

# Stages that can stop independently

## Concrete attempts before the invariant

**K:** triangle versus branch, then two-leaf path versus three-leaf star, give contrasting successful and stuck walks. Demonstrate resetting used-edge markers before another start. The two clean star diagrams make repairs genuinely different trials. Ask about leaving a leaf only after children have met one. Their own small map is discretionary; no degree vocabulary is necessary for the endpoint argument.

**Middle:** page 1 gives routes and several house starts before page 2 requests degree counts. A failed route is recorded as an attempt, not a proof. At an internal visit, put two fingers on arrival/departure lines and repeat if the route returns to that dot. Page 3 gives two distinct repairs (AB and AC), with fresh maps and space for routes. Then join the remaining odd dots to make a closed version. Page 4 designs on supplied dots; preserve the child's solution as well as the map. If numbering roads is easier than a letter list, use it.

**Upper:** actual attempts precede parity. Page 2 splices a loop at A and then at an internal B visit, so insertion is not just a slogan or an abstract algorithm. Page 3 explicitly changes from every road once to every road at least once, allowing repeats. Try two routes before counting the extra copies. Page 4 makes three concrete pair repairs, then changes the endpoint requirement. Page 5 compares two six-odd maps: the count alone gives the same lower bound, but one attains it and one does not.

## Proof reserves and extra

For Euler necessity, repeated visits still pair arrivals and departures; only endpoints can have one unpaired incident road. The two splices motivate, but do not prove, all-even sufficiency. Page 2 preserves the connectedness, finite-edge, leftover-even and temporary-edge arguments. For delivery, match a concrete route with a parity lower bound. Do not equate "six odd dots" with "three repeats suffice." The star is the counterexample.

The extra first totals actual weighted routes, then distinguishes a required long road from a cheap extra connection. Walk A–D–C to compare with AC; B–A–D to compare with BD. All original roads still require service. Only after that comparison introduce the three pairings. AD and CD tie for cheapest first pair but lead to different totals: greedy tie choice is not guaranteed safe. The augmentation decomposition proof on page 3 is an optional adult discussion after the child draws their own extras. Preserve shortest-path costs, not direct-road-only pairing costs.

## Record actual learning

Save route lists and failed attempts, which rule needed rescue, whether paired fingers or strips helped, and which challenges children wanted to continue. Distinguish observations from hypotheses, construction from optimality proof, and explained results from supplied theorems. Record each actually attempted v3 page; the rest remain unpiloted. The exact cases/sequence are our Week 1-informed adaptation, not claims about prior students' behavior.

# Additional v3 solution checks

## K and middle

K triangle: A-B-C-A or B-C-A-B. Branch: D-A-B-C-A; B cannot be the endpoint of a complete route because it has even degree while A,D are odd. The two-edge path A-B-C works; the three-leaf star cannot. With BC added, use D-A-B-C-A. With BD added, use C-A-B-D-A. Each ends at the center and unjoined leaf. Child maps need direct checking, not one assumed answer.

Middle house attempts from A cannot finish every edge once; C and D work as opposite endpoints. The branch degrees are A=3,B=2,C=2,D=1. K4 degrees all equal 3. House degrees and route are on page 2. With duplicate AB, a trail is C-A-B-D-A-B-C-D. With duplicate AC, a trail is B-A-C-B-D-A-C-D. Add CD in the first repair, or BD in the second, to make all degrees even. Closed routes appear below. For the five-dot design, the existing house is one valid answer; adding CD makes every degree even. Many designs work.

## Upper construction and costs

Problem 3: A-D-E-A-B-C-A. Problem 4: A-D-E-A-B-F-G-B-C-A. The degree at B changes from 2 to 4; all 9 roads appear once. Problem 5 optimum 8 and lower bound are on page 3. For that route, AB and CD are crossed twice; others once; every crossing-degree is 4. One extra copy fixes two of four odd degrees, so cannot make a closed walk possible.

Problem 8 closed routes: AB+CD uses A-B-C-D-A-C-D-B-A; AC+BD uses A-C-B-D-A-B-D-C-A; AD+BC uses A-D-B-C-A-B-C-D-A. Each costs 8. Problem 9 can duplicate CD, using A-B-C-A-D-C-D-B, length 7 with endpoints A,B. Problem 10 can add AD,BE,CF between triangles ABC and DEF and repeat those same three links; the 12-step route on page 3 uses every road. Problem 11's five-leaf star requires all five roads twice: A-B-A-C-A-D-A-E-A-F-A, length 10, five repeats. Each leaf requires both arrival and departure on its only road.

## Extra route certificates

Problem 1: any complete route is admissible; optimal total is 23. Problem 2 original sum 20. Cheapest connections and all pairing totals are on page 3. Problem 5's best AB+CD route is A-B-C-D-A-C-D-B-A, total 23. AD+BC can use A-D-B-C-A-B-C-D-A, total 24. Choosing AD first forces BC, adding  $1+3=4$ ; choosing CD first leaves AB, adding  $1+2=3$ . This refutes an unconditional cheapest-first guarantee, not every greedy run. Problem 7 for the optimum leaves separate AB and CD copies; their four endpoints are the original odd vertices. General augmentation pairing, including multi-road paths and nonnegative-cost cycles, is proved on page 3.